Bayesian Density Field Reconstruction and Two-Point Statistics from Log-Voronoi Gaussian Process Regression

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Abstract

We present GP-log ρ, a Bayesian framework for reconstructing the cosmological density field and its statistics from a discrete tracer catalogue. The key observation is that the logarithm of a Voronoi cell volume is an approximately Gaussian, unbiased estimator of the negative log-density at the tracer position, with a noise variance $\sigma_V^2 \approx 0.289$ derivable analytically from Poisson–Voronoi statistics. This converts density field inference into standard Gaussian Process (GP) regression: the posterior over the log-density field is closed-form, the two-point correlation function $\xi_\delta(r) = e^{K(r)} - 1$ is obtained directly from the inferred GP kernel $K(r)$, and all higher N-point functions are determined by $K(r)$ under the log-normal model. No MCMC is required. Computational cost is $\mathcal{O}(Nk^3)$ via Vecchia approximations implemented in the GraphGP/JAX library, making the method applicable to surveys with $N \sim 10^7$ tracers. Linear bias cancels exactly in log-space, leaving the inferred $K(r)$ and hence $\xi_\delta(r)$ independent of the linear bias factor. Tidal and second-order Lagrangian bias couple to the traceless tidal shear s^2, which has randomly oriented eigenvectors; this contaminates the monopole $K_0(r)$ but not the quadrupole $K_2^{(s)}(r)$, making the growth rate $f_{\rm growth}$ inferred from the redshift-space quadrupole immune to tidal bias. Per-point redshift-space distortions enter as an exact non-linear relation between redshift-space and real-space Voronoi volumes, yielding a single-snapshot estimator of $\nabla \cdot \mathbf{v}$ and $f_{\rm growth}$ without perturbation theory. A kNN shape tensor provides a model-free per-point estimate of the local tidal field and survey completeness simultaneously. We validate the framework on log-normal mocks and N-body simulations, and outline applications to GAMA, DESI, and the Quaia quasar catalogue. Super-sample covariance is rank-1 in configuration space and is automatically marginalised through the GP kernel amplitude posterior, requiring no external simulation suite. The inferred $K(r)$ simultaneously certifies an upper bound of $O(K(r)^2)$ on non-Gaussian covariance excess at each scale; at BAO scales this is $O(10^{-3})$ of the Gaussian contribution, making simulation-based covariance matrices unnecessary for the target scales of this framework.

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1 Introduction

The spatial distribution of galaxies and dark matter tracers encodes the full history of gravitational structure formation. Extracting that information requires reconstructing a continuous density field from a discrete, noisy point sample and then computing summary statistics—most fundamentally the two-point correlation function $\xi_\delta(r)$ and its Fourier transform, the power spectrum $P(k)$—together with reliable uncertainties.

Standard estimators such as Landy & Szalay (1993) measure ξ_δ by counting pairs in radial bins and comparing to a random catalog. They are unbiased and well understood, but they treat each scale independently, discard the smooth structure of the underlying field, and produce uncertainty estimates (jackknife or covariance-matrix inversion) that are decoupled from the density-field model. The complementary approach of kernel density estimation (KDE) and its cosmological refinement, the Delaunay Tessellation Field Estimator [DTFE; Schaap and van de Weygaert, 2000], reconstructs the density field adaptively from the point distribution but provides no uncertainty and no direct connection to two-point statistics.

A more principled framework treats the density field as a random field to be inferred from data. The Log-Gaussian Cox Process [LGCP; Møller et al., 1998] places a Gaussian process (GP) prior on $\log \rho(\mathbf{x})$ and models the observed points as a Poisson process with intensity $\rho(\mathbf{x})$. This is statistically correct but computationally expensive: the Poisson likelihood is non-conjugate to the GP prior, requiring Markov chain Monte Carlo or variational inference. In the cosmological setting, Jasche and Kitaura [2010] implement precisely this log-normal Poissonian posterior at the field level using Hamiltonian Monte Carlo, and the BORG algorithm [Jasche and Wandelt, 2013] extends it to infer the *initial conditions* of structure formation via a physical forward model. These are powerful approaches, but they address a harder problem than we consider here: HADES and BORG reconstruct the full dynamical history of the Universe; their non-conjugate likelihoods are unavoidable given that goal. Our aim is more focused—*optimal inference of the statistics of the final, observed density field*—and this permits a qualitatively simpler treatment.

The key observation is that the Voronoi tessellation of the tracer positions already contains the density information in a form that is nearly Gaussian. The Voronoi cell volume V_i around particle i satisfies $\hat{\rho}_i = 1/V_i$ (in units of mean density), so $\log V_i \approx -\log \rho(\mathbf{x}_i)$. For a Poisson point process the log-volume fluctuations are approximately Gaussian with a known variance $\sigma_V^2 \approx 0.29$ [from the Poisson–Voronoi distribution; Ferenc and Néda, 2007]. This converts the intractable Poisson likelihood into a linear Gaussian observation model, making the full GP posterior available in closed form.

In this paper we develop and validate this *GP*-$\log \rho$ framework. Its main properties are:

1. The posterior density field is available in closed form—no MCMC required—at cost $\mathcal{O}(Nk^3)$ using sparse Vecchia approximations [GraphGP; Dodge et al., 2025].

2. The GP kernel $K(r) = \xi_{\log \rho}(r)$ is inferred jointly with the field via marginal likelihood maximisation, and yields the matter two-point function through $\xi_\delta(r) = e^{K(r)} - 1$ [Coles and Jones, 1991] with a full posterior covariance across all scales simultaneously.

3. All N-point functions of ρ follow analytically from $K(r)$ under the log-normal model, so the computational cost of extracting higher-order information is no greater than fitting the two-point function.

4. The Hessian of the log-density field is available analytically from the GP posterior, enabling probabilistic cosmic web classification with calibrated uncertainties.

5. Biased tracers enter trivially: linear bias b shifts each observation by $-\log b$, leaving the inferred $K(r)$ unaffected.

The paper is organized as follows. Section 2 introduces the GP-$\log\rho$ model and derives the closed-form posterior. Section 3 develops the connection to $\xi_\delta(r)$ and compares to standard estimators. Section 4 treats higher-order statistics and computational scaling. Section 5 discusses biased and weighted tracers. Section 6 derives the Hessian field and its use for cosmic web classification. Section 7 validates the method on N-body simulations. Section 9 develops the exact nonlinear connection between Voronoi volumes, the Lagrangian Jacobian, and the velocity divergence field, and places RSD into the framework without perturbation theory. Section 8 outlines application to current surveys including GAMA, DESI, and the Quaia quasar catalog. Appendices provide detailed derivations of the Voronoi noise model (Appendix A), the Jensen bound (Appendix B), and the position-uncertainty extension (Appendix C).

2 The GP-$\log\rho$ Model

2.1 Prior: Gaussian process on the log-density

Let $\{\mathbf{x}_i\}_{i=1}^N$ be the positions of N tracers in a periodic box of volume L^3. Define the log-density contrast field

$$f(\mathbf{x}) \;\equiv\; \log\rho(\mathbf{x}) - \log\bar{\rho}, \tag{1}$$

where $\bar{\rho} = N/L^3$ is the mean density. We place a zero-mean Gaussian process prior on f,

$$f \;\sim\; \mathcal{GP}\big(0,\, K(\mathbf{x},\mathbf{x}')\big), \tag{2}$$

with stationary, isotropic kernel $K(r) \equiv K(|\mathbf{x}-\mathbf{x}'|)$. This induces a log-normal prior on $\rho(\mathbf{x}) = \bar{\rho}\, e^{f(\mathbf{x})}$, ensuring $\rho > 0$ everywhere. The kernel $K(r)$ is the two-point correlation function of the log-density field, $K(r) = \langle f(\mathbf{x}) f(\mathbf{x}+\mathbf{r})\rangle$, and relates to the matter correlation function through

$$\xi_\delta(r) \;=\; e^{K(r)} - 1, \tag{3}$$

a standard result for log-normal fields [Coles and Jones, 1991, Kayo et al., 2001].

Any positive-definite, isotropic kernel is a valid choice for $K(r)$. We consider the Matérn family as a concrete example,

$$M_\nu(r) \;=\; \sigma_f^2\, \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\, r}{\ell}\right)^{\nu} \mathcal{K}_\nu\!\left(\frac{\sqrt{2\nu}\, r}{\ell}\right), \tag{4}$$

with variance σ_f^2, length scale ℓ, and smoothness ν; here $\mathcal{K}_\nu$ denotes the modified Bessel function of the second kind, which we write with a calligraphic $\mathcal{K}$ to distinguish it from the GP kernel $K(r)$ throughout this paper. as well as a spectral-mixture kernel [Wilson and Adams, 2013] that can represent arbitrary stationary covariance structure. The matter power spectrum is recovered via $P(k) = \int K(r) e^{-i\mathbf{k}\cdot\mathbf{r}} d^3r$ applied to the inferred $K(r)$.

Crucially, *no compensation constraint* is required on $K(r)$. In the complementary GP-δ approach, the Gaussian prior on $\delta(\mathbf{x})$ must satisfy $\int \xi_\delta(r)\, 4\pi r^2\, dr = 0$ (volume

conservation), which forces Mexican-hat or engineered kernels. In the GP-$\log\rho$ model, volume conservation is guaranteed geometrically: the Voronoi cells partition L^3 exactly, so $\sum_i V_i = L^3$ identically regardless of the choice of $K(r)$. The compensation relation $\int \xi_\delta \, 4\pi r^2 \, dr = 0$ then emerges as a consequence of the data fit via equation (3), with the turnover scale and negative-lobe amplitude genuinely inferred rather than imposed.

2.2 Observation model: log-Voronoi volumes

Construct the Voronoi tessellation of the tracer positions. Each tracer i is assigned a cell volume V_i, the region of L^3 closer to $\mathbf{x}_i$ than to any other tracer. The natural density estimator is $\hat{\rho}_i = \bar{\rho}/V_i$, or equivalently

$$s_i \;\equiv\; \log V_i \;\approx\; -f(\mathbf{x}_i) \;+\; \epsilon_i, \tag{5}$$

where ϵ_i is noise arising from the discreteness of the point sample. For a locally Poisson point process, the distribution of $\log V_i$ at fixed underlying density is well approximated by a Gaussian with mean $-f(\mathbf{x}_i)$ and variance σ_V^2 [Ferenc and Néda, 2007]; see Appendix A for a derivation. Numerically,

$$\sigma_V^2 \;\approx\; 0.2887, \qquad c_V \;\equiv\; \langle \log V \rangle - \langle -f \rangle \;\approx\; 0.4228, \tag{6}$$

where c_V is a small mean offset absorbed into the GP mean.

Equation (5) is a *linear* observation model: observed $\mathbf{s} = (s_1, \ldots, s_N)$ are noisy linear measurements of the latent field values $\mathbf{f} = (f(\mathbf{x}_1), \ldots, f(\mathbf{x}_N))$,

$$\mathbf{s} \;=\; -\mathbf{f} \;+\; \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \;\sim\; \mathcal{N}(\mathbf{0},\, \sigma_V^2 \mathbf{I}). \tag{7}$$

This is the key simplification over the LGCP model, where the Poisson likelihood $p(\{x_i\} \mid \rho) = \exp(-\int \rho \, d^3x) \prod_i \rho(x_i)$ is non-conjugate to the GP prior. The Gaussian approximation in equation (7) is accurate when the GP kernel enforces smoothness on scales larger than the mean inter-particle spacing $n^{-1/3}$; we bound the approximation error in Appendix B.

2.3 Closed-form posterior

With the linear Gaussian model of equation (7), the joint distribution of $\mathbf{f}$ and $\mathbf{s}$ is Gaussian, and the posterior $p(\mathbf{f} \mid \mathbf{s})$ is available in closed form. Let $\mathbf{K}_{NN}$ denote the $N \times N$ matrix with entries $K_{ij} = K(|\mathbf{x}_i - \mathbf{x}_j|)$. Then

$$p(\mathbf{f} \mid \mathbf{s}) = \mathcal{N}(\boldsymbol{\mu}_f,\, \boldsymbol{\Sigma}_f), \tag{8}$$

$$\boldsymbol{\mu}_f = -\mathbf{K}_{NN}\big(\mathbf{K}_{NN} + \sigma_V^2 \mathbf{I}\big)^{-1}\mathbf{s}, \tag{9}$$

$$\boldsymbol{\Sigma}_f = \mathbf{K}_{NN} - \mathbf{K}_{NN}\big(\mathbf{K}_{NN} + \sigma_V^2 \mathbf{I}\big)^{-1}\mathbf{K}_{NN}. \tag{10}$$

Predictions at arbitrary test points $\mathbf{x}_*$ follow from standard GP conditioning:

$$\mu_f(\mathbf{x}_*) = -\mathbf{k}_*^T\big(\mathbf{K}_{NN} + \sigma_V^2 \mathbf{I}\big)^{-1}\mathbf{s}, \tag{11}$$

$$\sigma_f^2(\mathbf{x}_*) = K(0) - \mathbf{k}_*^T\big(\mathbf{K}_{NN} + \sigma_V^2 \mathbf{I}\big)^{-1}\mathbf{k}_*, \tag{12}$$

where $\mathbf{k}_*$ has entries $K(|\mathbf{x}_* - \mathbf{x}_i|)$. The reconstructed density field and its uncertainty are then $\hat{\rho}(\mathbf{x}_*) = \bar{\rho}\, e^{\mu_f(\mathbf{x}_*)}$ and $\sigma_{\log\rho}(\mathbf{x}_*) = \sigma_f(\mathbf{x}_*)$.

2.4 Alternative observation model: kNN volumes and the Hamilton estimator

The exact Voronoi tessellation can be replaced by k-nearest-neighbour (kNN) volume estimates $\hat{V}_i^{(k)} = (4\pi/3)r_{k,i}^3/k$, which are unbiased estimators of $1/\lambda_i$ with known noise variance $\sigma_k^2 = \psi_1(k)$ (trigamma function). The partition identity $\sum_i V_i = L^3$ is restored by rescaling as in equation (67). Appendix D derives the full noise model and quantifies the information content on $K(r)$ as a function of scale; in particular, $k = 1$ is the self-consistent choice given the position uncertainty analysis of Appendix C.

For observational catalogs that provide a random catalog encoding the survey selection function, the data/random distance ratio

$$\hat{s}_i^{\mathrm{H}} = 3 \log \frac{r_{k_R,i}^{(R)}}{r_{k_D,i}^{(D)}} + \log \frac{N_R}{N_D} \tag{13}$$

cancels the selection function exactly at the per-point level (Appendix E), with noise variance $\psi_1(k_D) + \psi_1(k_R)$. This is the per-point, field-level analogue of the Hamilton (1993) estimator. The two KD-trees required to evaluate it are the same trees used for any two-point estimator, so the incremental computational cost is negligible.

2.5 Kernel hyperparameter inference

The kernel hyperparameters $\boldsymbol{\theta} = \{\sigma_f^2, \ell, \ldots\}$ are inferred by maximising the log marginal likelihood,

$$\log p(\mathbf{s} \mid \boldsymbol{\theta}) \;=\; -\tfrac{1}{2}\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s} - \tfrac{1}{2}\log|\mathbf{C}| - \tfrac{N}{2}\log 2\pi, \tag{14}$$

where $\mathbf{C} = \mathbf{K}_{NN} + \sigma_V^2\mathbf{I}$ is the total covariance. The gradient $\partial \log p/\partial\boldsymbol{\theta}$ is available analytically and optimisation is performed by L-BFGS with multiple random restarts. The marginal likelihood is non-convex in $\boldsymbol{\theta}$: local minima corresponding to spurious short-length-scale fits occur when $\ell \ll \bar{d}$ (the inter-particle spacing), but are strongly penalised by the $-\frac{1}{2}\log|\mathbf{C}|$ term, which diverges as $\ell \to 0$. In practice, initialising ℓ near $\bar{d}$ and σ_f^2 near the sample variance of $\{s_i\}$ places the optimiser in the basin of the physical solution. For spectral-mixture kernels with multiple components, we use 5–10 restarts and select the solution with highest marginal likelihood. Because the marginal likelihood integrates out the field values, it directly measures how well the kernel explains the spatial structure of the log-volume data; its maximum over $\boldsymbol{\theta}$ simultaneously determines the best-fit kernel $K(r)$, which is the log-density correlation function and therefore encodes $\xi_\delta(r)$ via equation (3).

2.6 Computational scaling via Vecchia approximations

Exact GP inference requires $\mathcal{O}(N^3)$ operations, prohibitive for $N \gtrsim 10^4$. We use the Vecchia approximation [Vecchia, 1988], in which each $f(\mathbf{x}_i)$ is conditioned only on its k nearest previously-ordered neighbours. This sparsifies $\mathbf{C}^{-1}$ to a banded Cholesky factor, reducing cost to $\mathcal{O}(Nk^3)$. The GraphGP library [Dodge et al., 2025] implements this in JAX with automatic differentiation, enabling GPU acceleration and gradient-based kernel optimisation at scale. For $k = 30$ the approximation error is negligible relative to the Voronoi noise floor σ_V^2.

3 Two-Point Statistics

3.1 The kernel as the two-point function

The log marginal likelihood (14) is maximised over the kernel hyperparameters $\boldsymbol{\theta}$, so the inferred $K(r;\hat{\boldsymbol{\theta}})$ is the maximum-likelihood estimate of the log-density correlation function $\xi_{\log\rho}(r)$. The matter two-point function follows immediately:

$$\hat{\xi}_\delta(r) \;=\; \exp\bigl[K(r;\hat{\boldsymbol{\theta}})\bigr] - 1. \tag{15}$$

This is not a post-processing step applied to a separately estimated density field; it is the *direct output* of kernel optimisation. The kernel $K(r)$ and the field reconstruction are inferred simultaneously, without a separate pair-counting step.

3.2 Posterior uncertainty on $\xi_\delta(r)$

The posterior distribution over $K(r)$ arises from uncertainty in the hyperparameters $\boldsymbol{\theta}$. In the Laplace approximation around the maximum-likelihood point, the posterior covariance of the kernel evaluated at a set of lags $\{r_1,\ldots,r_m\}$ is

$$\mathrm{Cov}\bigl[K(r_a),K(r_b)\bigr] \;\approx\; \left(\frac{\partial K(r_a)}{\partial\boldsymbol{\theta}}\right)^T \mathbf{H}^{-1}\left(\frac{\partial K(r_b)}{\partial\boldsymbol{\theta}}\right), \tag{16}$$

where $\mathbf{H} = -\partial^2 \log p(\mathbf{s}\mid\boldsymbol{\theta})/\partial\boldsymbol{\theta}^2$ is the Hessian of the log marginal likelihood. Propagating through equation (15) via the delta method,

$$\mathrm{Cov}\bigl[\hat{\xi}_\delta(r_a),\hat{\xi}_\delta(r_b)\bigr] \;\approx\; e^{K(r_a)+K(r_b)}\,\mathrm{Cov}\bigl[K(r_a),K(r_b)\bigr]. \tag{17}$$

This covariance matrix over the full function $\xi_\delta(r)$ is produced automatically as a by-product of optimisation; no jackknife resampling or covariance matrix estimation from mocks is required.

Three features distinguish this uncertainty estimate from standard approaches. First, it is *field-level*: uncertainty arises from the global fit of $K(r)$ to all spatial correlations simultaneously, not from noise in individual pair counts. Second, it captures *off-diagonal covariance* between different scales, which is missed by bin-by-bin estimators. Third, shot noise enters through the known per-point variance σ_V^2, which is cleanly separated from the signal kernel $K(r)$; in Landy–Szalay, the shot-noise subtraction is a global $1/N$ correction that assumes spatial uniformity.

3.3 Band-power representation and $P(k)$

For a non-parametric representation, we expand the kernel in band powers, $K(r) = \sum_j A_j\phi_j(r)$, where $\phi_j(r) = j_0(k_j r)$ are spherical Bessel functions at wavenumbers k_j. The optimisation over $\{A_j\}$ then directly infers the power spectrum of the log-density field; the matter power spectrum follows by Fourier transforming $\xi_\delta(r) = e^{K(r)} - 1$. The Fisher matrix on $\{A_j\}$ from the Hessian of the marginal likelihood is the analogue of the CMB band-power window-function matrix, derived here from the actual posterior geometry rather than assumed Gaussian statistics.

3.4 Comparison to Landy–Szalay

Before the comparison, one important caveat on survey geometry. The claim that GP-$\log\rho$ avoids a random catalog requires qualification. Voronoi cells near a survey boundary extend into the unobserved region and inflate V_i spuriously; those cells must be clipped to the survey geometry, which requires the same angular mask and selection function knowledge as the LS random catalog. The genuine advantage is that this geometric information is encoded *once*, at the tessellation stage, after which the GP marginal likelihood handles the window function automatically without pair counting. The need to characterise the survey geometry is not avoided; the algorithmic cost of incorporating it is reduced.

Table 1 summarises the remaining differences.

Table 1: Comparison of two-point estimators.

Property	GP-$\log\rho$	Landy–Szalay
Uncertainty estimate	Full posterior covariance matrix	Jackknife or mock-based
Scale coupling	Captured	Ignored
Shot noise	Per-point σ_V^2, separated	Global $1/N$ subtraction
Survey geometry	Encoded at tessellation stage	Random catalog + pair counting
Density field	Joint posterior reconstruction	Not produced
Hessian / web type	Analytic from posterior	Not available
Cost	$\mathcal{O}(Nk^3)$	$\mathcal{O}(N^2)$ pair counting

The key trade-off is that GP-$\log\rho$ relies on the log-normal model for ρ, which is accurate in the weakly and mildly non-linear regimes but imperfect deep inside collapsed structures. For the scales relevant to BAO and the linear power spectrum, this is an excellent approximation [Kayo et al., 2001]. On strongly non-linear scales ($r \lesssim 1\,h^{-1}$Mpc), the log-normal model underestimates the skewness of the one-point density PDF, and the LS estimator is less model-dependent. Both regimes can be validated against simulations; Section 7 quantifies the accuracy.

3.5 Super-sample covariance and the non-Gaussian covariance bound

SSC in configuration space is rank-1. A super-survey background mode δ_b shifts the survey mean density $\bar{\rho} \to \bar{\rho}(1+\delta_b)$, which in log-density space is a spatially constant subtraction $f(\mathbf{x}) \to f(\mathbf{x}) - \log(1+\delta_b)$ applied uniformly to all s_i. After subtracting the sample mean this leaves a scale-independent additive constant $(\delta_b^{(\log)})^2$ added to $K(r)$ at all separations. The SSC contribution to the covariance of ξ_δ is therefore

$$\mathrm{Cov}_{\mathrm{SSC}}(\xi_\delta(r_1),\, \xi_\delta(r_2)) \;=\; \big(1+\xi_\delta(r_1)\big)\big(1+\xi_\delta(r_2)\big)\,\sigma_b^2(V), \tag{18}$$

which is *exactly rank-1* in (r_1, r_2) space. No scale-dependent response function is required — a simplification unique to the configuration-space / log-density basis, in contrast to Fourier space where $\partial P(k)/\partial\delta_b$ requires perturbation theory or separate-universe simulations.

The rank-1 structure is automatically captured in the GP posterior. The background mode is degenerate with the DC component of $K(r)$, i.e. the kernel amplitude $\sigma_f^2 = K(0)$. Marginalising over σ_f^2 during kernel hyperparameter inference propagates $\sigma_b^2(V)$ directly

into the posterior covariance $\mathrm{Cov}_{\mathrm{post}}(K(r_1), K(r_2))$ as its rank-1 component, with no external SSC calculation. The integral constraint — the requirement that $\hat{\xi}_\delta(r)$ integrate to zero over the survey volume — is a corollary: the GP zero-mean condition and marginalisation over σ_f^2 handle it internally, with the integral constraint uncertainty correctly reflected in the posterior rather than applied as a post-hoc correction.

Self-certified non-Gaussian covariance bound. In the log-normal model all connected n-point functions are determined by $K(r)$. The connected trispectrum contribution to the covariance of ξ_δ — the dominant non-Gaussian term beyond SSC — is, at leading order in K,

$$\Delta\mathrm{Cov}_{\mathrm{nG}}(\xi_\delta(r_1), \xi_\delta(r_2)) \;\sim\; O\big(K(r_1)^2\, K(r_2)\big) \;\sim\; \xi_\delta(r_1)^2\, \xi_\delta(r_2). \tag{19}$$

The ratio of this non-Gaussian excess to the Gaussian (mode-counting) covariance at separation r is therefore $O(K(r))$: at BAO scales $K(r) \sim 10^{-3}$, the non-Gaussian excess is three orders of magnitude below the Gaussian covariance; at intermediate scales $r \sim 30\,h^{-1}\mathrm{Mpc}$, $K(r) \sim 0.05$, bounding the excess at 5%.

The key point is that $K(r)$ is *measured* from the same data and the same inference. Our posterior on $K(r)$ therefore simultaneously constitutes a *self-certified upper bound* on the non-Gaussian covariance excess at every scale: if the inferred $K(r_{\mathrm{large}})$ is small and the log-normal residuals are well-characterised in the GP, the non-log-normal trispectrum excess is bounded by $O(K^3)$ without any simulation input. Simulations provide genuinely new information only on scales where $K(r)$ is large — but these are precisely the nonlinear scales where the log-normal model is already imperfect for independent reasons.

Together, the automatic SSC marginalisation and the self-certified non-Gaussian bound provide a complete covariance treatment at BAO and larger scales, at no additional computational cost beyond the GP inference itself.

4 Higher-Order Statistics

4.1 N-point functions from the log-normal model

Under the log-normal model, the density field $\rho(\mathbf{x}) = \bar{\rho}\, e^{f(\mathbf{x})}$ with $f \sim \mathcal{GP}(0, K)$, all N-point functions of ρ are determined entirely by the two-point function $K(r)$ of the Gaussian field f. This follows from the moment-generating function of the multivariate normal: for any set of positions $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$,

$$\left\langle \prod_{i=1}^{N} \rho(\mathbf{x}_i) \right\rangle \;=\; \bar{\rho}^N \exp\left[\tfrac{1}{2} \sum_{i,j} K(|\mathbf{x}_i - \mathbf{x}_j|) \right]. \tag{20}$$

In particular, the three-point function (bispectrum) and four-point function (trispectrum) are given by

$$\xi_3(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \bar{\rho}^3 e^{K_{12}+K_{13}+K_{23}+K_{11}/2+K_{22}/2+K_{33}/2} - \text{lower-order terms}, \tag{21}$$

$$\xi_4(\ldots) = \ldots, \tag{22}$$

where $K_{ij} \equiv K(|\mathbf{x}_i - \mathbf{x}_j|)$ and all connected parts factor through K. The Isserlis (Wick) theorem applies to the Gaussian field f; the non-Gaussianity of ρ is entirely encoded in the exponential mapping.

The practical consequence is that, once $K(r)$ is inferred from the GP marginal likelihood as described in Section 2, the full hierarchy of N-point functions is determined. There is no additional computational cost associated with going from the two-point to the N-point function. In particular:

- The bispectrum $B(k_1, k_2, k_3)$ follows by Fourier transforming the connected three-point function derived from equation (21).
- The counts-in-cells (CiC) statistics—the probability distribution of the number of tracers in a randomly placed sphere of radius R—are determined by integrating the N-point functions over spherical volumes, which reduces to integrals over $K(r)$.
- The void probability function (VPF) $P_0(R)$, the probability that a sphere of radius R is empty, can be computed from the same integrals.

4.2 Posterior predictive distribution of the density

At any point $\mathbf{x}_*$, the GP posterior gives a Gaussian distribution over $f(\mathbf{x}_*)$ with mean μ_* and variance σ_*^2 (equations 11–12). The posterior distribution of the density at that point is therefore log-normal:

$$p(\rho(\mathbf{x}_*) \mid \mathbf{s}) \;=\; \mathrm{LogNormal}\big(\bar{\rho}\, e^{\mu_*},\, \sigma_*^2\big)\,. \tag{23}$$

This gives immediate access to credible intervals on $\rho(\mathbf{x}_*)$ and to the posterior predictive density PDF across the entire volume.

4.3 Computational cost

A key advantage is that the $\mathcal{O}(Nk^3)$ cost of the GP fit is incurred once, after which all derived statistics are cheap to evaluate. Compare this to direct N-point estimators, which scale as $\mathcal{O}(N^n)$ for the n-point function. For a survey of $N = 10^6$ galaxies and $n = 3$, direct counting requires $\mathcal{O}(10^{18})$ operations; with $k = 30$ in the Vecchia approximation, the GP fit requires $\mathcal{O}(30^3 \times 10^6) \approx 3 \times 10^{10}$ operations, and all higher-order statistics follow analytically.

This efficiency rests on the log-normal model. The model encodes a specific functional form for higher-order statistics; it is not model-independent. In practice, the log-normal model is accurate for the mildly non-linear regime ($\delta \lesssim$ few) that dominates most of the cosmological signal in galaxy surveys. Section 7 quantifies where deviations from log-normality become significant.

4.4 Connection to k-nearest-neighbour statistics

The k-nearest-neighbour cumulative distribution function (kNN-CDF) [Banerjee and Abel, 2021] is a powerful non-Gaussian summary statistic. Under the log-normal model, the kNN-CDF can be expressed as a functional of $K(r)$ by integrating the void probability function over spherical shells. This provides a direct analytic link between the GP kernel inferred from log-Voronoi observations and the kNN statistics, suggesting a unified framework for the two approaches [Gangopadhyay et al., 2025]. We defer a detailed treatment to future work.

5 Biased Tracers

Galaxy surveys do not observe the dark matter density directly; they observe luminous tracers that are biased with respect to the underlying matter field. A central virtue of the GP-$\log \rho$ framework is that deterministic linear bias enters as a trivial additive shift of the observations, leaving the inference of $K(r)$ unchanged.

5.1 Linear deterministic bias

Suppose the galaxy number density is related to the matter density by $\rho_g(\mathbf{x}) = b\,\rho_m(\mathbf{x})$, where $b > 0$ is a constant bias factor. Then the log-density contrast of galaxies is

$$f_g(\mathbf{x}) = \log \rho_g - \log \bar{\rho}_g = \log \rho_m - \log \bar{\rho}_m = f_m(\mathbf{x}), \tag{24}$$

since the $\log b$ terms cancel. The Voronoi volumes V_i are computed from the galaxy positions, so the observations are $s_i = \log V_i \approx -f_g(\mathbf{x}_i) + \epsilon_i = -f_m(\mathbf{x}_i) + \epsilon_i$.

The observation model is therefore *identical* to the unbiased case. The inferred kernel $K(r)$ is the log-density correlation function of the matter field, $\xi_{\log \rho_m}(r)$, and $\xi_\delta(r) = e^{K(r)} - 1$ is the matter two-point function, regardless of b. This exact cancellation of linear bias in log-space is a key advantage over the GP-δ approach, where b modifies the likelihood function in a non-trivial way.

5.2 Scale-dependent bias

For scale-dependent bias $b(k)$, the galaxy log-density is

$$f_g(\mathbf{x}) = f_m * h(\mathbf{x}) + \eta(\mathbf{x}), \tag{25}$$

where h is a filter with Fourier transform $\hat{h}(k) = \log b(k)/\log \bar{b}$ (schematically) and η is stochastic noise. The effective kernel for the galaxy observations becomes

$$K_g(r) = \int \frac{d^3k}{(2\pi)^3}\, |\hat{h}(k)|^2 \hat{K}_m(k)\, e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{26}$$

which can be modelled by choosing a spectral-mixture kernel whose shape is jointly optimised with a bias transfer function $\hat{h}(k)$. For surveys where scale-dependent bias is known from simulations (e.g. through halo occupation distribution modelling), $\hat{h}(k)$ can be fixed a priori; otherwise it can be marginalised over as additional kernel hyperparameters.

5.3 Stochastic bias and Poisson shot noise

Beyond deterministic bias, tracers carry stochastic shot noise: galaxies are a point process whose overdispersion relative to Poisson sampling contributes extra variance to the Voronoi log-volumes. The effective per-point noise becomes

$$\sigma^2_{\mathrm{eff},i} = \sigma^2_V + \sigma^2_{\mathrm{stoch}}, \tag{27}$$

where $\sigma^2_{\mathrm{stoch}}$ encodes stochasticity beyond Poisson (sub-Poisson for regular lattices, super-Poisson for clustered tracers). If $\sigma^2_{\mathrm{stoch}}$ is uniform across the sample, it enters as an additional additive contribution to the diagonal noise variance and is jointly inferred with the kernel hyperparameters. Its value is degenerate with the small-scale amplitude of $K(r)$; a smoothness prior or a minimum scale cut breaks the degeneracy.

5.4 Multi-tracer analysis

When multiple tracer populations (e.g. emission-line galaxies and luminous red galaxies from DESI) are available in the same volume, they can be combined in a joint GP with a shared matter kernel $K_m(r)$ but population-specific noise levels $\sigma^2_{V,\alpha}$ and stochastic bias amplitudes $\sigma^2_{\text{stoch},\alpha}$. The joint marginal likelihood is

$$\log p(\{\mathbf{s}_\alpha\} \mid K_m, \{\sigma_\alpha\}) = \sum_\alpha \log p(\mathbf{s}_\alpha \mid K_m, \sigma_\alpha), \tag{28}$$

where the sum over populations α is exact because the observation noises are independent. Multi-tracer analysis suppresses cosmic variance on large scales [Seljak, 2009], and the GP framework inherits this suppression naturally through the shared kernel constraint.

5.5 Magnitude- and luminosity-weighted tracers

A tracer of luminosity L_i (or apparent magnitude m_i) can be assigned an effective weight w_i reflecting its contribution to a weighted density field $\rho_w(\mathbf{x}) = \sum_i w_i \delta^{(3)}(\mathbf{x} - \mathbf{x}_i)$. In log-space, weighting shifts the observation by $-\log w_i$:

$$s_i \to s_i - \log w_i, \tag{29}$$

which is again a simple additive modification. FKP weighting [Feldman et al., 1994] for optimal power-spectrum estimation enters in the same way, with $w_i = 1/(1 + \bar{n}P_0)$ evaluated at each point's local density estimate.

5.6 Lagrangian bias expansion, tidal operators, and joint inference

Operator basis. The complete Lagrangian bias expansion to second order is [McDonald and Roy, 2009, Senatore, 2015, Desjacques et al., 2018]

$$\delta_g^L(\mathbf{q}) \;=\; b_1^L\,\delta_L + b_2^L\,\delta_L^2 + b_{s^2}^L\,s^2 + b_{t^3}^L\,t^3 + \ldots, \tag{30}$$

where $\delta_L = -DI_1$ is the linear density contrast, $s^2 = s_{ij}s_{ij} \propto I_1^2 - 3I_2$ is the traceless tidal shear squared, and $t^3 \propto I_1^3 - \frac{9}{2}I_1I_2 + \frac{27}{2}I_3$ is the cubic tidal operator. The symmetric invariants $I_1 = \sum_a \lambda_a$, $I_2 = \sum_{a<b} \lambda_a\lambda_b$, $I_3 = \prod_a \lambda_a$ of the deformation eigenvalues are precisely the quantities controlling the Jacobian anisotropy of Appendix F. The Lagrangian bias expansion and the Voronoi anisotropy correction are polynomials in the same basis.

Log-Jacobian as a bias expansion. Expanding $\log J = \sum_a \log(1 - \lambda_a D)$,

$$\log J \;=\; -D\,I_1 - \tfrac{D^2}{2}(I_1^2 - 2I_2) - \tfrac{D^3}{3}(I_1^3 - 3I_1I_2 + 3I_3) + \mathcal{O}(D^4), \tag{31}$$

with coefficients fixed by LPT. The observation $s_i = -\log J + \ldots$ is already an expansion in bias operators. For biased tracers,

$$s_i^{(g)} \;=\; -\log J(\mathbf{q}_i) \;-\; c_V(\boldsymbol{\lambda}_i) \;+\; b_1^L\,\delta_{L,i} \;+\; b_{s^2}^L\,s_i^2 \;+\; b_{t^3}^L\,t_i^3 \;+\; \epsilon_i. \tag{32}$$

As shown in Section 5, b_1^L cancels exactly in log-space. The tidal terms $b_{s^2}^L$ and $b_{t^3}^L$ do *not* cancel and must be inferred jointly with $K(r)$.

Three anisotropic effects with distinct symmetry axes. With real redshift-space observations, three physically distinct effects produce anisotropy in the per-point data:

1. *RSD* (Section 9): shifts s_i by $\partial_\parallel v_\parallel / H$, with preferred axis fixed at $\hat{\mathbf{e}}_\parallel$ (line of sight), constant across the survey.

2. *Tidal bias* $b^L_{s^2} s_i^2$: preferred axis aligned with the local deformation eigenvectors, varying with the LSS environment.

3. *Voronoi anisotropy* $c_V(\boldsymbol{\lambda}_i)$: same axis as the tidal field (it is a function of the same deformation eigenvalues), but a deterministic function of the GP Hessian of f rather than a free parameter.

All three are entangled in the scalar s_i alone but are separable because they carry different orientational signatures relative to the line of sight.

The shape tensor as a second observation channel. The traceless data-minus-random shape tensor $\Delta\widetilde{\mathbf{T}}_i \equiv \mathbf{T}_i^{(D)} - \mathbf{T}_i^{(R)} - \frac{1}{3}\mathbf{I}$ is a direct noisy per-point measurement of the local tidal field. It decomposes as

$$\Delta\widetilde{\mathbf{T}}_i \;=\; \alpha_{\rm RSD}\,\mathbf{P}_{\parallel,i} \;+\; \beta\big(b^L_{s^2} + \gamma_V(\boldsymbol{\lambda}_i)\big)\,\tilde{s}_{ij}(\mathbf{q}_i) \;+\; \boldsymbol{\epsilon}_i^{(T)}, \tag{33}$$

where $\mathbf{P}_\parallel = \hat{\mathbf{e}}_\parallel \otimes \hat{\mathbf{e}}_\parallel - \frac{1}{3}\mathbf{I}$ is the line-of-sight projector (fixed orientation), $\tilde{s}_{ij}$ is the traceless tidal shear (varying orientation), $\alpha_{\rm RSD} \propto f_{\rm growth}$, β is a geometric calibration constant, γ_V encodes the Voronoi-to-tidal coupling from Appendix F, and $\boldsymbol{\epsilon}_i^{(T)}$ is angular sampling noise with variance $\sim 1/K$ per component for K neighbours. Because $\Delta\widetilde{\mathbf{T}}_i$ is built from neighbour *directions* only, its noise channel is independent of ϵ_i in s_i. The shape tensor is therefore not merely a diagnostic but a second independent observable, to be entered jointly with s_i into the GP marginal likelihood.

Separability by orientation statistics. The RSD and tidal contributions in equation (33) have qualitatively different orientation statistics across the survey. The RSD term carries a *fixed* axis $\hat{\mathbf{e}}_\parallel$ that survives averaging over survey volume; the tidal term carries a *varying* axis that averages to zero over uncorrelated environments. No model assumption is needed for this separation:

- Project $\Delta\widetilde{\mathbf{T}}_i$ onto $\hat{\mathbf{e}}_\parallel$ to extract $\alpha_{\rm RSD} \propto f_{\rm growth}$ (fixed-axis component).

- Cross-correlate the residual with the GP Hessian estimate of $\tilde{s}_{ij}$ to extract $b^L_{s^2} + \gamma_V$ (varying-axis component).

- Compute γ_V from the GP posterior Hessian via Appendix F and subtract to isolate $b^L_{s^2}$.

The Voronoi correction γ_V and the physical tidal bias $b^L_{s^2}$ remain degenerate within the varying-axis component, broken only by the fact that γ_V is a *deterministic* function of the GP posterior whereas $b^L_{s^2}$ is a free parameter. The Doroshkevich prior (Appendix F) regularises the eigenvalue posterior and prevents over-fitting this decomposition in weakly anisotropic environments.

Full joint observation model. The complete per-point observation vector for a biased tracer in redshift space is

$$\mathbf{y}_i = \begin{pmatrix} s_i \\ \Delta\widetilde{\mathbf{T}}_i \end{pmatrix} = \begin{pmatrix} -f_i - c_V(\boldsymbol{\lambda}_i) + b_{s^2}^L s_i^2 + \log|1 + \partial_\parallel v_{\parallel,i}/H| \\ \alpha_{\rm RSD}\,\mathbf{P}_{\parallel,i} + \beta\big(b_{s^2}^L + \gamma_V(\boldsymbol{\lambda}_i)\big)\tilde{s}_{ij}(\mathbf{q}_i) \end{pmatrix} + \begin{pmatrix} \epsilon_i \\ \boldsymbol{\epsilon}_i^{(T)} \end{pmatrix}, \tag{34}$$

with $\epsilon_i \perp \boldsymbol{\epsilon}_i^{(T)}$. The two rows share the signal $\tilde{s}_{ij}$, introducing a cross-channel correlation that enables joint inference of $f_{\rm growth}$, $b_{s^2}^L$, and $K(r)$ from the same dataset. Beyond the kernel, the free parameters are $f_{\rm growth}$ (from $\alpha_{\rm RSD}$), $b_{s^2}^L$ (tidal bias), and the Voronoi calibration coefficients $\{c_V^{(2)}, \sigma_V^{2,(2)}\}$, the latter pinnable from simulations.

Connection to the GP kernel. Tidal bias $b_{s^2}^L$ breaks the isotropy of the effective kernel: the residual covariance of s_i acquires an angular dependence $K(\mathbf{r}) = K_0(r) + b_{s^2}^L\, K_2(r)\, P_2(\hat{\mathbf{r}}\cdot\hat{\mathbf{n}}) + \ldots$, where $\hat{\mathbf{n}}$ is the local tidal axis [Desjacques et al., 2018]. Fitting this anisotropic kernel simultaneously infers $K_0(r)$ and $b_{s^2}^L$ at field level, with the shape tensor channel providing the independent constraint that prevents the two from trading off in the scalar s_i channel alone.

6 Hessian Field and Cosmic Web Classification

The morphology of the cosmic web—knots, filaments, walls, and voids—is encoded in the local Hessian of the density field. The GP-$\log\rho$ posterior provides the Hessian analytically, together with full uncertainty quantification.

6.1 Hessian of the log-density GP

Since $f(\mathbf{x})$ is a Gaussian process with kernel $K(r)$, its spatial derivatives are also Gaussian processes. The posterior mean and covariance of the gradient and Hessian follow from differentiating the kernel:

$$\langle \partial_i f(\mathbf{x}_*)\rangle = \partial_{x_{*,i}}\,\mu_f(\mathbf{x}_*), \tag{35}$$

$$\langle \partial_i\partial_j f(\mathbf{x}_*)\rangle = \partial_{x_{*,i}}\partial_{x_{*,j}}\,\mu_f(\mathbf{x}_*), \tag{36}$$

where $\mu_f(\mathbf{x}_*)$ is the posterior mean from equation (11). The kernel of the Hessian components is

$$\mathrm{Cov}[\partial_i\partial_j f(\mathbf{x}),\, \partial_k\partial_l f(\mathbf{x}')] = \partial_{x_i}\partial_{x_j}\partial_{x'_k}\partial_{x'_l} K(|\mathbf{x}-\mathbf{x}'|), \tag{37}$$

which is computable analytically for Matérn and RBF kernels. The posterior Hessian at any point is therefore a 3×3 Gaussian random matrix with known mean and covariance.

6.2 Chain-rule Hessian of the density

The physical Hessian of the density field $\rho = \bar{\rho}\, e^f$ is related to the log-density Hessian by the chain rule:

$$\partial_i\partial_j\rho = \rho\big(\partial_i\partial_j f + \partial_i f\,\partial_j f\big). \tag{38}$$

The extra rank-1 outer-product term $(\nabla f)(\nabla f)^T$ is absent in the GP-δ approach, where one differentiates the density contrast directly. This term amplifies curvature at density peaks (where $|\nabla f|$ is locally small but the second-order structure is enhanced by the exponential) and is important for the correct identification of filament axes and node positions.

6.3 T-web classification

The T-web scheme [Hahn et al., 2007] classifies each point $\mathbf{x}$ by the number of positive eigenvalues of the tidal tensor, which in our context is approximated by the Hessian of the density field (or the log-density field, as a computationally cheaper variant). Let $\lambda_1 \geq \lambda_2 \geq \lambda_3$ be the eigenvalues of $\partial_i \partial_j f(\mathbf{x}_*)$ from the GP posterior. The web type is:

Web type	Condition
Void	$\lambda_1 < \lambda_{\rm th}$
Wall	$\lambda_1 \geq \lambda_{\rm th} > \lambda_2$
Filament	$\lambda_2 \geq \lambda_{\rm th} > \lambda_3$
Knot	$\lambda_3 \geq \lambda_{\rm th}$

where $\lambda_{\rm th}$ is a threshold tuned to the mean density level.

Because the posterior Hessian is a Gaussian random matrix, the distribution of eigenvalues at each point is tractable. We draw N_s samples from the posterior Hessian distribution at $\mathbf{x}_*$ and classify each; the fraction of samples assigned to each web type gives the posterior probability $p(\text{web type} \mid \mathbf{s})$. This probabilistic classification is not available from DTFE or other single-estimate methods, where the web type is assigned deterministically.

6.4 V-web and velocity-field connection

The V-web [Hoffman et al., 2012] classifies the cosmic web through the velocity shear tensor rather than the density Hessian. In linear theory, $\mathbf{v}(\mathbf{k}) = i f H \, \mathbf{k}/k^2 \, \delta(\mathbf{k})$, so the velocity shear is related to $\nabla^{-2} \partial_i \partial_j \delta$. From the inferred density field and its Hessian, we can compute the linear-theory velocity shear at any point by applying the appropriate Green's function in the periodic box. This provides a V-web classification from the same GP posterior, at no additional data requirement.

6.5 Comparison to DTFE-based classification

DTFE computes a single density estimate and Hessian at each point; the web classification is deterministic and carries no uncertainty. The GP framework produces a posterior probability distribution over web types, which properly propagates the noise from finite sampling. In underdense regions (voids), where the Voronoi volumes are large and σ_V^2 per unit density is higher, the GP uncertainty on the Hessian eigenvalues is larger, and the posterior probability of "void" is spread over a range of threshold values. This reflects genuine epistemic uncertainty rather than a spurious single-valued estimate, and has direct consequences for void statistics and void galaxy cross-correlations.

7 Validation on Simulations

We validate the GP-$\log \rho$ framework on controlled simulations where the ground truth is known exactly, proceeding from idealised log-normal mocks to full N-body catalogs.

7.1 Simulation setup

Log-normal mocks. We generate Gaussian random fields $f(\mathbf{x})$ on a 256^3 periodic mesh with a prescribed input kernel $K_{\rm in}(r)$, derived from a ΛCDM linear power spectrum at

$z = 0$ via $K_{\rm in}(r) = \log(1 + \xi_\delta^{\rm lin}(r))$. Point samples of $N = 10^4$–10^6 are drawn from a Poisson process with intensity $\rho(\mathbf{x}) = \bar{\rho}\, e^{f(\mathbf{x})}$. These mocks test the inference machinery in the regime where the model is exact.

N-body catalogs. We apply the method to dark matter particle snapshots and halo catalogs from a suite of $L = 500\,h^{-1}$Mpc, 512^3-particle simulations run with GADGET-4 [Springel et al., 2021] at $z = 0, 0.5, 1$. These test the log-normal model assumption and quantify deviations on non-linear scales.

Implementation. The Voronoi tessellation is computed with VORO++ [Rycroft, 2009]. The GP inference and kernel optimisation use the GraphGP library in JAX with the Vecchia approximation at $k = 30$ nearest neighbours. Optimisation uses L-BFGS for 200–500 iterations.

7.2 Density field recovery

Figure **??** (placeholder) shows the posterior mean $\hat{f}(\mathbf{x})$ versus the true log-density field for a 10^5-particle log-normal mock in a $2D$ slice.

Coverage tests. At each of 10^4 test points, we record whether the true $f(\mathbf{x})$ falls within the 68% and 95% posterior credible intervals from equations (11)–(12). For log-normal mocks, coverage should be exact. For N-body outputs, we quantify the miscoverage on non-linear scales as a function of local density.

Resolution and scale cuts. Below the mean inter-particle spacing $\bar{d} = n^{-1/3}$, the Voronoi noise floor dominates and the posterior collapses to the prior. Meaningful density reconstruction requires $\ell \gtrsim \bar{d}$; the marginal likelihood automatically enforces this by penalising kernels with $\ell < \bar{d}$.

7.3 Two-point function recovery

Log-normal mocks. We compare the inferred $\hat{\xi}_\delta(r) = e^{\hat{K}(r)} - 1$ to the input $\xi_\delta^{\rm in}(r)$. On scales $r > 2\bar{d}$, the recovery is unbiased to within the posterior uncertainty. We compare the width of the GP posterior credible band on $\xi_\delta(r)$ to the jackknife uncertainty from Landy–Szalay on the same mock, finding comparable statistical power with the GP estimator capturing off-diagonal scale covariance that jackknife misses.

N-body catalogs. The log-normal approximation $\xi_\delta = e^K - 1$ underestimates the true ξ_δ on scales $r \lesssim 5\,h^{-1}$Mpc at $z = 0$ due to non-Gaussian skewness in the density PDF. On scales $r > 10\,h^{-1}$Mpc, including the BAO feature, the recovery is accurate to within 2%.

Beyond log-normal: theoretical avenues. The log-normal model is the simplest member of a family of transform-Gaussian field models. Two extensions are immediately available within the GP framework without altering the closed-form inference structure.

Non-stationary noise. On strongly non-linear scales the per-point noise $\sigma_{V,i}^2$ becomes density-dependent (overdense regions have denser tessellations and smaller cells with

lower relative noise; underdense regions show the opposite). Replacing $\sigma_V^2\mathbf{I}$ with a diagonal heteroscedastic noise model $\mathrm{diag}\{\sigma_{V,i}^2(f_i)\}$, evaluated at the current GP posterior mean in a fixed-point iteration (Appendix C), partially corrects for this and extends accurate field reconstruction to scales $r \sim 2\,h^{-1}$Mpc without departing from the log-normal prior.

Non-stationary kernel. The log-normal model assumes a stationary kernel $K(r)$, i.e. the same two-point structure everywhere. In practice, dense regions (halos) and underdense regions (voids) have distinct small-scale clustering. A locally adaptive length scale $\ell(\mathbf{x}) \propto n(\mathbf{x})^{-1/3}$ (the local inter-particle spacing), implementable via the input-warping trick [Snoek et al., 2014], allows the kernel to track the local non-linear scale. The Vecchia approximation continues to apply since neighbour sets are defined geometrically.

A fully non-Gaussian forward model (e.g. replacing the log-normal prior with a learned normalising flow, as in BORG-NF) would break the closed-form posterior and require variational or MCMC inference. Such models recover accuracy on strongly non-linear scales at the cost of $\sim 100\times$ greater computation. For applications targeting BAO and f_{growth} the log-normal model is adequate; for void profiles and one-point PDF constraints, the heteroscedastic noise extension suffices; the full non-Gaussian model is deferred to future work.

7.4 Hessian and web classification

We compare the GP-inferred web classification (posterior mode) to the reference T-web classification computed from the N-body density field on a fine mesh. The confusion matrix (knot/filament/wall/void vs reference) is presented for several values of the threshold λ_{th}. We report the fraction of mass in each web type and compare to the reference fractions. The probabilistic classification provides meaningful uncertainty estimates in transition regions (wall–filament boundaries) where single-estimate methods assign web type based on noise-dominated Hessian eigenvalue ratios.

7.5 Computational scaling

Table 2 reports wall-clock time for GP inference as a function of N and Vecchia parameter k on a single A100 GPU. The $\mathcal{O}(Nk^3)$ scaling is confirmed for $N = 10^4$–5×10^6. For $N = 10^6$ and $k = 30$, total inference time (Voronoi tessellation + kernel optimisation + posterior prediction on a 256^3 mesh) is approximately 20 minutes.

Table 2: Indicative wall-clock times (A100 GPU, $k = 30$).

N	Voronoi	GP optimise	Predict (256^3)
10^4	$< 1\,$s	$\sim 30\,$s	$\sim 2\,$min
10^5	$\sim 5\,$s	$\sim 5\,$min	$\sim 5\,$min
10^6	$\sim 1\,$min	$\sim 15\,$min	$\sim 5\,$min

8 Application to Survey Data

We outline the systematic considerations specific to each survey and the modifications to the baseline GP-$\log\rho$ model they require.

8.1 Survey geometry and selection function

Real surveys introduce two classes of systematic that must be handled before applying the GP:

1. *Angular mask and holes*: regions of zero or reduced completeness due to bright stars, detector gaps, or fibre collisions. Tracers near mask edges have Voronoi cells that extend into masked regions, inflating V_i spuriously. We clip Voronoi cells at the survey boundary using the convex hull or a randoms-based mask, and flag cells with boundary-fraction exceeding a threshold.

2. *Radial selection function*: the mean number density $\bar{n}(z)$ varies with redshift due to the flux limit and target selection. We work with the overdensity $f_i = \log(\rho_i/\bar{\rho}(z_i))$, subtracting the expected log-mean density at each redshift from the Voronoi log-volume: $s_i \to s_i - \log[\bar{n}(z_i)/\bar{n}]$.

8.2 GAMA

The Galaxy And Mass Assembly [GAMA; Driver et al., 2022] survey provides a highly complete ($> 98\%$), flux-limited spectroscopic sample at $z < 0.5$ over $\sim 250\,\mathrm{deg}^2$. Its small angular footprint and well-characterised selection make it the natural first real-data target for the GP-$\log\rho$ method. The Voronoi Volume Function (VVF) of the GAMA equatorial fields was studied by Paranjape and Alam [2020] as a summary statistic; the GP framework extends this to a full field-level inference with $\xi_\delta(r)$ uncertainty.

8.3 DESI

The Dark Energy Spectroscopic Instrument [DESI; DESI Collaboration et al., 2016] provides four distinct tracer populations (BGS, LRG, ELG, QSO) with independent bias parameters over $0.1 < z < 3.5$ and $\sim 14{,}000\,\mathrm{deg}^2$. The multi-tracer formalism of Section 5 applies directly: each population has its own σ_V^2 and stochastic noise, but shares the matter kernel $K_m(r)$. The joint inference suppresses cosmic variance on scales larger than the survey correlation length [Seljak, 2009]. Fibre-assignment incompleteness is handled via per-galaxy weights provided in the DESI value-added catalogs; these enter as $s_i \to s_i - \log w_i$ per equation (5).

8.4 Quaia

The Quaia catalog [Storey-Fisher et al., 2024] cross-matches *Gaia* DR3 astrometry with unWISE photometry to produce $\sim 1.3 \times 10^6$ quasars at $z \lesssim 4$, distributed over the extragalactic sky. Quaia carries a distinct and more complex set of systematics than the galaxy surveys above:

- *Gaia scanning law*: the *Gaia* detection completeness is a strongly anisotropic, non-monotonic function of sky position, driven by the satellite's precessing scan pattern. The resulting angular selection function has correlated structure on degree scales that mimics large-scale clustering if uncorrected.

- *Stellar contamination and star-quasar separation*: the colour-selection boundary in *Gaia*–unWISE space varies with stellar density, introducing a spatially varying purity that anti-correlates with the Galactic plane.

- *Dust extinction*: *Gaia* optical magnitudes are affected by Galactic dust extinction, with spatial structure correlated with the large-scale structure signal at low Galactic latitudes. The dust map itself carries uncertainty [e.g. Edenhofer et al., 2024], which should be propagated.
- *Wide redshift distribution*: Quaia spans a broad photometric redshift range ($\Delta z \sim 0.5$ per bin) requiring careful treatment of the radial selection function and projection effects.

These systematics differ qualitatively from flux-limited galaxy surveys (GAMA, DESI) and from the other quasar catalogs used for large-scale structure [e.g. eBOSS; Alam et al., 2021]. The GP-$\log\rho$ model accommodates them through a position-dependent effective weight $w_i(\hat{n}_i) = C(\hat{n}_i)/C_{\max}$, where $C(\hat{n})$ is the local completeness estimated from the systematics maps, so that $s_i \to s_i - \log w_i$. The GP kernel is then inferred from the corrected observations; residual systematics enter as an additional contribution to $\sigma^2_{\rm eff}$ and are jointly fitted. The Quaia cross-correlation with CMB lensing [Farren et al., 2024] provides an independent validation of the recovered density field amplitude.

9 Redshift-Space Distortions and the Lagrangian Density

The log-Voronoi observation model is not merely a convenient density estimator: it is the logarithm of the Jacobian of the Lagrangian-to-Eulerian coordinate map. This identification establishes a deep connection between the GP-$\log\rho$ framework and the velocity divergence field, and places RSD into the formalism exactly, without perturbation theory.

9.1 The Voronoi volume as the Lagrangian Jacobian

Let $\mathbf{q}$ denote the Lagrangian (initial) position of a fluid element and $\mathbf{x} = \mathbf{q} + \mathbf{\Psi}(\mathbf{q})$ its Eulerian position. By mass conservation, the density field satisfies

$$\rho(\mathbf{x})\, J(\mathbf{q}) \;=\; \bar{\rho}, \qquad J \;\equiv\; \det\!\left(\frac{\partial \mathbf{x}}{\partial \mathbf{q}}\right), \tag{39}$$

so $J(\mathbf{q}_i) = e^{-f(\mathbf{x}_i)}$, exactly. The Voronoi cell volume V_i is the Eulerian volume of the region closer to $\mathbf{x}_i$ than to any other tracer; in the continuum limit it converges to $1/n(\mathbf{x}_i) = \bar{n}^{-1}\, e^{-f(\mathbf{x}_i)}$. Therefore

$$\log V_i \;=\; -f(\mathbf{x}_i) + \text{const} \;=\; \log J(\mathbf{q}_i) + \text{const}. \tag{40}$$

The log-Voronoi observation is not an estimator of the Jacobian—it *is* the Jacobian. The GP prior on $f(\mathbf{x})$ is therefore a prior on $\log J(\mathbf{q})$, and the posterior on f is a posterior on the local volume stretching of the Lagrangian map.

9.2 Velocity divergence as the material derivative of $\log J$

The Eulerian continuity equation, $\partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0$, can be written as a material derivative:

$$\frac{d\log\rho}{dt} + \nabla\cdot\mathbf{v} \;=\; 0 \quad\Longrightarrow\quad \frac{d\log J}{dt} \;=\; \nabla\cdot\mathbf{v}. \tag{41}$$

This is an exact nonlinear statement. The velocity divergence field is the time derivative of the log Voronoi volume evaluated at the particle. No linearisation, no perturbation theory, no grid assignment is required.

Implication for simulations. For two snapshots of an N-body simulation at times t and $t + \Delta t$, the per-particle velocity divergence estimate is

$$(\nabla \cdot \mathbf{v})_i \;\approx\; \frac{\log V_i(t + \Delta t) - \log V_i(t)}{\Delta t}, \tag{42}$$

with the same Voronoi (or kNN) noise model at each snapshot. No velocity field interpolation, no finite-difference grid, and no kernel smoothing is required. The GP-$\log \rho$ framework applied to pairs of snapshots therefore provides a full field-level estimate of $\nabla \cdot \mathbf{v}$ with calibrated uncertainty.

Linear theory connection. In linear perturbation theory, $\mathbf{\Psi} = -\nabla_{\mathbf{q}} \Phi_0 / D$ and $\nabla_{\mathbf{q}} \cdot \mathbf{\Psi} = -\delta$, so $\nabla \cdot \mathbf{v} = -H f_{\text{growth}}\, \delta$, where $f_{\text{growth}} = d \log D / d \log a$ is the logarithmic growth rate. Equation (41) at linear order therefore gives

$$\frac{d \log V_i}{dt} \;\approx\; -H f_{\text{growth}}\, \delta_i, \tag{43}$$

recovering the standard result that the divergence of the peculiar velocity traces the (negative) density contrast, weighted by the growth rate.

9.3 The logarithmic derivative of the Lagrangian density

There is a further identity worth making explicit. Define the Lagrangian density as the density of a fluid element expressed in its initial coordinates:

$$\rho_L(\mathbf{q}) \;\equiv\; \rho(\mathbf{x}(\mathbf{q}))\, J(\mathbf{q}) \;=\; \bar{\rho}, \tag{44}$$

which is constant by mass conservation. Taking the logarithmic derivative with respect to the displacement amplitude D (treating the displacement field as $D\boldsymbol{\psi}(\mathbf{q})$ in LPT),

$$\frac{d \log \rho_L}{dD} \;=\; 0 \quad \Longrightarrow \quad \frac{d \log J}{dD} \;=\; -\frac{d \log \rho}{dD}. \tag{45}$$

The logarithmic derivative of J with respect to the growth factor is the negative logarithmic derivative of the Eulerian density. In the GP-$\log \rho$ framework, both sides are directly accessible: $\log J = -f$ is the inferred log-density field, and its derivative with respect to D (estimated from two-snapshot comparisons or from linear-theory forward modelling) gives $f_{\text{growth}}\, \delta(\mathbf{x})$ per point. This closes the loop between the density field inference and the velocity field without requiring a separate peculiar-velocity measurement.

9.4 Redshift-space Voronoi volumes

Galaxy surveys measure positions in *redshift space*: the observed coordinate along the line of sight $\hat{\mathbf{e}}_\parallel$ is

$$s_\parallel \;=\; x_\parallel + \frac{v_\parallel}{H}, \tag{46}$$

while transverse coordinates are unchanged. The Jacobian of the redshift-space mapping is

$$J^{(s)}(\mathbf{q}) \;=\; J(\mathbf{q})\left|1+\frac{1}{H}\frac{\partial v_\parallel}{\partial x_\parallel}\right|, \tag{47}$$

so the redshift-space log-Voronoi volume satisfies

$$\log V_i^{(s)} \;=\; \log V_i \;+\; \log\left|1+\frac{\partial v_\parallel}{H\partial x_\parallel}\right|_i \;\approx\; \log V_i \;+\; \left.\frac{(\nabla\cdot\mathbf{v})_\parallel}{H}\right|_i, \tag{48}$$

where the approximation holds when the velocity gradient is sub-Hubble. The *difference* $\log V_i^{(s)} - \log V_i$ is exactly the line-of-sight velocity gradient per unit H — nonlinearly, without any moment expansion or multipole decomposition.

Single-snapshot inference. In practice we observe only $V_i^{(s)}$, not V_i. The GP-$\log\rho$ model in redshift space therefore infers the *redshift-space* log-density field $f^{(s)}(\mathbf{s})$, whose correlation function is the anisotropic kernel $K^{(s)}(r_\perp, r_\parallel)$. In linear theory,

$$K^{(s)}(r_\perp, r_\parallel) \;=\; K(r)\left(1+f_{\rm growth}\mu^2\right)^2 \;+\; \mathcal{O}(\delta^3), \tag{49}$$

where $\mu = r_\parallel/r$ is the cosine of the angle to the line of sight and $r^2 = r_\perp^2 + r_\parallel^2$. The GP marginal likelihood optimisation over an anisotropic kernel $K^{(s)}(r_\perp, r_\parallel)$ therefore simultaneously infers the real-space $K(r)$ and the growth rate $f_{\rm growth}$, with a full posterior covariance between them. This is the field-level analogue of fitting the power-spectrum multipoles P_0, P_2, P_4.

9.5 RSD from the shape tensor

The anisotropy of the redshift-space Voronoi tessellation is already encoded in the data shape tensor $\mathbf{T}_i^{(D)}$ defined in Appendix D. In redshift space, $\mathbf{T}_i^{(D)}$ is compressed along $\hat{\mathbf{e}}_\parallel$ in overdense regions (Kaiser squashing) and elongated in very dense regions (Fingers of God). The random shape tensor $\mathbf{T}_i^{(R)}$ is isotropic in the survey interior (randoms have no clustering). Their difference,

$$\Delta\mathbf{T}_i \;\equiv\; \mathbf{T}_i^{(D)} - \mathbf{T}_i^{(R)}, \tag{50}$$

is a per-point estimator of the local velocity shear anisotropy, corrected for survey geometry via $\mathbf{T}_i^{(R)}$. Specifically, the projection of $\Delta\mathbf{T}_i$ onto $\hat{\mathbf{e}}_\parallel$ measures the line-of-sight compression/elongation at point i, providing a per-point estimate of $\partial v_\parallel/\partial x_\parallel$ without pair counting, multipole decomposition, or velocity reconstruction.

Connection to the V-web. The eigenstructure of $\Delta\mathbf{T}_i$ identifies the local velocity shear axes, which are the V-web classification directions of Section 6. In the linear regime, $\Delta\mathbf{T}_i \propto -f_{\rm growth}\,\mathbf{H}_i^f$ (the Hessian of f), recovering the V-web result from kinematic data alone. In the nonlinear regime the two differ, providing a direct nonlinear diagnostic of the departure from linear theory at each point.

9.6 Summary of connections

The identification $\log V_i = \log J(\mathbf{q}_i) + \text{const}$ links four otherwise separate threads:

1. *Density field*: $f = -\log J$, directly from Voronoi volumes.
2. *Velocity divergence*: $\nabla \cdot \mathbf{v} = d\log J/dt$, from Voronoi volume evolution between snapshots.
3. *Growth rate*: $f_{\text{growth}} = -(\nabla \cdot \mathbf{v})/(H\delta)$, from the ratio of the velocity divergence to the density field, both inferred from the same GP posterior.
4. *RSD*: $\log V^{(s)} - \log V = (\nabla \cdot \mathbf{v})_{\parallel}/H$, encoding the line-of-sight velocity gradient as an anisotropic correction to the log-Voronoi observations.

All four are accessible within the GP-$\log\rho$ framework, using only tracer positions and their Voronoi (or kNN) volumes. No velocity catalog, no moment expansion, and no grid-based density assignment is required.

10 Conclusions

We have presented a framework for Bayesian density field reconstruction and two-point statistics inference from galaxy and tracer point catalogs, based on a Gaussian process prior on the log-density field combined with a Gaussian observation model derived from Voronoi cell volumes. The central result is that the log-Voronoi observation model is conjugate to the GP prior, yielding a closed-form posterior at $\mathcal{O}(Nk^3)$ cost and converting an otherwise intractable log-Gaussian Cox process into a standard GP regression problem.

The main conclusions are as follows.

Closed-form posterior. The Voronoi log-volume $s_i = \log V_i$ is a linear, Gaussian observation of the log-density field $f(\mathbf{x}_i)$ with a known noise variance $\sigma_V^2 \approx 0.29$. This makes the full GP posterior available in closed form, without MCMC or variational approximations, at $\mathcal{O}(Nk^3)$ cost via Vecchia sparse approximations.

Joint field and two-point function inference. The GP kernel $K(r)$ is the log-density correlation function. Maximising the marginal likelihood over kernel hyperparameters jointly infers the density field and $\xi_\delta(r) = e^{K(r)} - 1$ with a full posterior covariance matrix across all scales. No separate pair-counting step, random catalog, or jackknife resampling is required.

Volume conservation without compensation. The discrete identity $\sum_i V_i = L^3$ encodes volume conservation geometrically. Any positive-definite kernel is admissible; the compensation constraint that must be imposed on GP-δ kernels emerges automatically from the data.

Higher-order statistics at no extra cost. Under the log-normal model, all N-point functions of ρ are determined by $K(r)$. Once the two-point function is inferred, the bispectrum, counts-in-cells, and void probability function are available analytically.

Biased and weighted tracers. Linear bias cancels exactly in log-space; the inferred $K(r)$ is the matter two-point function regardless of b. General bias, stochastic noise, and FKP weights enter as additive shifts of the per-point observations.

Lagrangian Jacobian identity and RSD. The log-Voronoi volume is the logarithm of the Lagrangian Jacobian $J(\mathbf{q}_i) = e^{-f(\mathbf{x}_i)}$ (Section 9). This identification gives four exact, nonlinear connections without perturbation theory: the density field ($f = -\log J$), the velocity divergence ($\nabla \cdot \mathbf{v} = d \log J/dt$, measurable from two-snapshot Voronoi volume ratios), the growth rate f_{growth} (from the ratio of velocity divergence to density), and RSD ($\log V^{(s)} - \log V = (\nabla \cdot \mathbf{v})_{\parallel}/H$, encoding the line-of-sight velocity gradient as anisotropic kernel distortion).

Probabilistic cosmic web classification. The GP posterior yields analytic mean and covariance of the density Hessian, enabling probabilistic T-web and V-web classification with calibrated uncertainty at each point. The chain-rule term $(\nabla f)(\nabla f)^T$ contributes a physically meaningful rank-1 correction to the Hessian of ρ that is absent in GP-δ treatments.

Survey applications. We have outlined application to GAMA, DESI, and the Quaia quasar catalog. Quaia presents a qualitatively different set of systematics (Gaia scanning law, stellar contamination, dust extinction) from flux-limited galaxy surveys, which the per-point weighting and heteroscedastic noise model accommodate naturally.

Super-sample covariance and non-Gaussian covariance bound. In configuration space, super-sample covariance (SSC) is a rank-1 contribution to $\mathrm{Cov}(\xi_\delta(r_1), \xi_\delta(r_2))$, factoring as $(1 + \xi_\delta(r_1))(1 + \xi_\delta(r_2))\sigma_b^2(V)$ (Section 3.5). This rank-1 structure is degenerate with the DC component of $K(r)$ and is automatically marginalised in the GP posterior — no separate SSC calculation or simulation suite is required. Furthermore, the non-Gaussian trispectrum excess above the Gaussian covariance is $O(K(r)^2)$ at each scale, and since $K(r)$ is measured from the same inference, the posterior on $K(r)$ constitutes a self-certified upper bound on non-Gaussian covariance contributions at every scale. On BAO scales this bound is $O(10^{-3})$ relative to the Gaussian covariance, rendering expensive simulation-based covariance matrices unnecessary for the scales this framework targets.

Outlook. The most immediate extensions are (i) full validation on N-body catalogs across redshift, (ii) application to GAMA with comparison of the inferred $\xi_\delta(r)$ to published Landy–Szalay results, and (iii) a Quaia cross-correlation with CMB lensing using the reconstructed density field. Longer-term, the GP marginal likelihood provides a natural route to simultaneous inference of $\xi_\delta(r)$, the bias model, and the cosmological parameters, by replacing the kernel hyperparameters with a parameterised ΛCDM power spectrum and optimising over cosmological parameters directly. At the Vecchia approximation cost of $\mathcal{O}(Nk^3)$, this is computationally accessible for current and forthcoming survey volumes.

A Voronoi Volume Distribution and Noise Calibration

A.1 Poisson–Voronoi distribution

For a stationary Poisson point process with intensity λ in $\mathbb{R}^3$, the distribution of Voronoi cell volumes has been studied analytically and numerically. Let $v = \lambda V$ be the dimensionless cell volume. The distribution $p(v)$ is well approximated by a generalised gamma distribution [Ferenc and Néda, 2007],

$$p(v) \;=\; \frac{(3897/2)^{31/2}}{\Gamma(31/2)}\, v^{29/2} \exp\!\left(-\frac{3897}{2}\, v\right), \tag{51}$$

which gives mean $\langle v \rangle = 1$, variance $\mathrm{Var}(v) = 2/3897 \approx 5.13 \times 10^{-4}$, and

$$\mathrm{Var}(\log v) \;=\; \psi_1(31/2) \;\approx\; 0.2887 \;\equiv\; \sigma_V^2, \tag{52}$$

where ψ_1 is the trigamma function. The mean log-volume is

$$\langle \log v \rangle \;=\; \psi_0(31/2) - \log(3897/2) \;\approx\; -0.4228 \;\equiv\; -c_V, \tag{53}$$

where ψ_0 is the digamma function. These are the numerical values quoted in equation (6) of the main text.

A.2 Gaussian approximation for $\log V$

From equations (51)–(53), the log-volume has mean $-c_V - f(\mathbf{x}_i)$ and variance σ_V^2 in the locally Poisson limit. The skewness and excess kurtosis of $\log v$ under the generalised gamma are small ($\gamma_1 \approx 0.06$, $\gamma_2 \approx 0.006$), justifying the Gaussian approximation $\log V_i \sim \mathcal{N}(-c_V - f(\mathbf{x}_i),\, \sigma_V^2)$. The residual non-Gaussianity is negligible compared to the signal variance $K(0) = \sigma_f^2$ for all cosmologically relevant power spectra.

A.3 Heteroscedastic noise from local density

The locally Poisson approximation assumes the point process is Poisson within each cell. In regions of elevated density, the Voronoi cells are small and the per-cell Poisson noise is larger relative to the cell volume. To first order, the effective noise variance scales as

$$\sigma_{V,i}^2 \;\approx\; \sigma_V^2 \left(1 + \left.\frac{\partial \log \sigma_V^2}{\partial \log \lambda}\right|_{\lambda=\bar{\lambda}} (e^{f_i} - 1)\right), \tag{54}$$

where the derivative can be estimated from simulations. This heteroscedastic correction is $\mathcal{O}(\delta)$ and is significant only in dense clusters. For the validation in Section 7 we use the uniform $\sigma_V^2 = 0.2887$ as the baseline and assess the impact of the correction on recovery statistics.

B Jensen Bound and Regime of Validity

B.1 Jensen correction to the mean

The density estimator $\hat{\rho}_i = \bar{\rho}/V_i$ satisfies $\mathbb{E}[\hat{\rho}_i] = \bar{\rho}\,\mathbb{E}[1/V_i]$, while the GP prior gives $\rho(\mathbf{x}_i) = \bar{\rho}\, e^{f_i}$. By Jensen's inequality applied to the convex function $\phi(v) = 1/v$, $\mathbb{E}[1/V_i] \geq$

$1/\mathbb{E}[V_i]$, so $\hat{\rho}_i$ is a slightly biased high estimator of $\rho(\mathbf{x}_i)$. In log-space, Jensen applied to the concave function log gives

$$\mathbb{E}[\log V_i \mid f_i] \;=\; -f_i - c_V + \delta_J, \tag{55}$$

where c_V is the mean log-volume of a unit-intensity Poisson Voronoi cell (equation 53) and δ_J is the Jensen residual from the variation of f *within* cell i.

B.2 Magnitude of the residual

The residual δ_J arises because the density is not exactly constant within a Voronoi cell; the cell volume responds to the *average* density near $\mathbf{x}_i$, not the exact value at $\mathbf{x}_i$. Expanding $f(\mathbf{x})$ in a Taylor series around $\mathbf{x}_i$ and integrating over the cell,

$$\delta_J \;\approx\; \frac{1}{2}\,\sigma^2_{\text{intra}}, \qquad \sigma^2_{\text{intra}} \;=\; \frac{1}{V_i}\int_{V_i} \big[f(\mathbf{x}) - f(\mathbf{x}_i)\big]^2\, d^3x. \tag{56}$$

For a kernel with length scale ℓ and variance σ_f^2,

$$\sigma^2_{\text{intra}} \;\sim\; \sigma_f^2\left(1 - e^{-r^2_{\text{cell}}/\ell^2}\right), \quad r_{\text{cell}} \sim n^{-1/3}, \tag{57}$$

where r_{cell} is the typical cell radius. The correction is therefore suppressed as $r_{\text{cell}}/\ell \to 0$, i.e. when the smoothing scale greatly exceeds the inter-particle spacing. For $\ell > 3\,n^{-1/3}$, we have $\sigma^2_{\text{intra}}/\sigma_f^2 < 0.01$ and δ_J is negligible compared to σ_V^2.

B.3 Practical criterion

The GP-$\log\rho$ approximation is valid when

$$\ell \;\gtrsim\; 3\,n^{-1/3}(\mathbf{x}) \;=\; 3\left(\bar{\rho}\,e^{f(\mathbf{x})}\right)^{-1/3}. \tag{58}$$

In overdense regions ($f > 0$), the local inter-particle spacing is smaller, so the requirement on ℓ is less stringent. In underdense regions, Voronoi cells are large and the intra-cell gradient of f can be non-negligible; here σ^2_{intra} is larger and a small δ_J correction should be applied. In practice, the marginal likelihood marginalises over f and automatically weights down predictions in regions where the approximation is poor (through the increased posterior variance), so the Jensen correction is a refinement rather than a necessity.

C Position Uncertainty Extension

C.1 Sources of effective position error

The observation model $s_i \approx -f(\mathbf{x}_i) + \epsilon_i$ assumes that the tracer position $\mathbf{x}_i$ is known exactly. Two effects introduce an effective uncertainty in the representative density position:

1. *Voronoi centroid offset*: the Voronoi cell is not centered on $\mathbf{x}_i$; the centre of mass of cell i is displaced from $\mathbf{x}_i$ by $\boldsymbol{\delta}_i$ with $|\boldsymbol{\delta}_i| \sim V_i^{1/3}/(d+2)^{1/2}$ in d dimensions.

2. *Spectroscopic/photometric redshift uncertainty*: line-of-sight position errors $\sigma_z/(1+z)\,c/H(z)$ can be significant for photometric surveys.

C.2 Heteroscedastic noise from centroid displacement

If the field gradient at $\mathbf{x}_i$ is $\mathbf{g}_i = \nabla f(\mathbf{x}_i)$, then the error in using $f(\mathbf{x}_i)$ as a proxy for the cell-averaged log-density is

$$\sigma^2_{\text{pos},i} \;\approx\; |\mathbf{g}_i|^2 \cdot \frac{V_i^{2/3}}{d+2} \;=\; |\nabla f(\mathbf{x}_i)|^2 \cdot \frac{V_i^{2/3}}{d+2}. \tag{59}$$

This contributes additively to the per-point noise:

$$\sigma^2_{\text{eff},i} \;=\; \sigma^2_V + \sigma^2_{\text{pos},i}. \tag{60}$$

Because $\sigma^2_{\text{pos},i}$ depends on the posterior gradient $\nabla f(\mathbf{x}_i)$, the noise model and the posterior are coupled. We handle this via a fixed-point iteration:

1. Initialise with $\sigma^2_{\text{eff},i} = \sigma^2_V$ for all i.
2. Compute the posterior mean μ_f and gradient $\hat{\mathbf{g}}_i = \nabla \mu_f(\mathbf{x}_i)$.
3. Update $\sigma^2_{\text{eff},i} = \sigma^2_V + |\hat{\mathbf{g}}_i|^2 V_i^{2/3}/(d+2)$.
4. Recompute the posterior with the updated noise.
5. Repeat until convergence (typically 3–5 iterations).

In smooth regions, $|\hat{\mathbf{g}}_i|$ is small and $\sigma^2_{\text{eff},i} \approx \sigma^2_V$. Near filaments and walls, the gradient is large and the effective noise is inflated, correctly reducing the influence of individual tracers in morphologically complex environments.

C.3 Adaptive kernel from positional uncertainty

An equivalent treatment replaces the fixed kernel $K(r)$ with a locally adaptive kernel that accounts for the effective smoothing imposed by position uncertainty. Define the per-point length scale $\Lambda_i = V_i^{2/3}/(d+2)$. The effective covariance between observations at $\mathbf{x}_i$ and $\mathbf{x}_j$ is

$$K_{\text{eff}}(i,j) \;=\; K\left(\sqrt{r_{ij}^2 + \tfrac{1}{2}(\Lambda_i + \Lambda_j)}\right), \tag{61}$$

which smears the kernel over the geometric mean of the two cells' effective radii. This adaptive kernel reduces to the original $K(r_{ij})$ in the limit of small Voronoi cells (dense sampling) and captures the increased uncertainty in underdense regions automatically.

Both the heteroscedastic noise approach (equation 60) and the adaptive kernel approach (equation 61) produce essentially equivalent results for moderate density contrasts; the latter is somewhat cheaper as it avoids the fixed-point iteration.

D Multi-Order kNN Volume Estimator

D.1 Single-order kNN estimator

For a Poisson point process with local intensity $\lambda_i = \bar{n}\, e^{f(\mathbf{x}_i)}$ at point i, the distance $r_{k,i}$ to the k-th nearest neighbour satisfies

$$u_{k,i} \;\equiv\; \lambda_i \tfrac{4\pi}{3} r_{k,i}^3 \;\sim\; \text{Gamma}(k,1). \tag{62}$$

The log-volume estimate $\hat{s}_i^{(k)} = 3\log r_{k,i} - \log(k/\bar{n}) + \log(4\pi/3)$ is a Gaussian observation of $-f(\mathbf{x}_i)$ with

$$\mathbb{E}[\hat{s}_i^{(k)}] = -f(\mathbf{x}_i) - c_k, \qquad c_k \equiv \log k - \psi_0(k), \tag{63}$$

$$\mathrm{Var}[\hat{s}_i^{(k)}] = \sigma_k^2 \equiv \psi_1(k). \tag{64}$$

Numerically: $\sigma_1^2 = \pi^2/6 \approx 1.645$, $\sigma_5^2 \approx 0.221$, $\sigma_{10}^2 \approx 0.105$, $\sigma_{20}^2 \approx 0.051$, compared with $\sigma_V^2 \approx 0.289$ for the exact Voronoi volume.

D.2 Self-consistency at $k = 1$ and the position uncertainty floor

There is a natural choice of order that follows from the position uncertainty analysis of Appendix C. The effective noise on the observation $\hat{s}_i$ has two contributions: density estimation error σ_k^2, and position uncertainty arising from the displacement of the Voronoi centroid from $\mathbf{x}_i$ (equation 59),

$$\sigma_{\mathrm{pos},i}^2 \;\approx\; |\nabla f(\mathbf{x}_i)|^2 \cdot \frac{V_i^{2/3}}{d+2}. \tag{65}$$

The characteristic cell radius entering equation (65) is $V_i^{1/3}$. For the kNN estimator at order $k = 1$, $V_i^{1/3} \sim r_{1,i}$ — the nearest-neighbour distance. But $r_{1,i}$ is precisely the maximum displacement of $\mathbf{x}_i$ before it enters another point's Voronoi territory: no point can be displaced by more than $r_{1,i}/2$ without leaving its own cell. The density noise σ_1^2 and the position uncertainty $\sigma_{\mathrm{pos}}^2 \sim |\nabla f|^2 r_{1,i}^2$ are therefore both governed by the same geometric quantity $r_{1,i}$; they are not independent error budgets but the same fact viewed from two angles.

Larger k reduces $\sigma_k^2 = \psi_1(k)$ below the position noise floor without reducing the position uncertainty itself (which depends on the actual point spacing, not on k). Precision gained in density estimation is then lost to the unresolved question of where exactly within the cell the underlying density $f(\mathbf{x}_i)$ should be evaluated. In this sense $k = 1$, the distance to the first *other* data point, is the self-consistent choice: density noise and position noise are matched.

The exact Voronoi estimator achieves $\sigma_V^2 \approx 0.289 < \sigma_1^2$ by exploiting the full polyhedral geometry of the cell — specifically, the face areas and normals — to average over the solid angle, not merely the circumradius. This is not a contradiction: the Voronoi estimator uses *directional* information (Section D.6) that is absent from the scalar $r_{1,i}$, and pays for the lower density noise with higher computational cost and boundary complexity.

D.3 Joint noise structure for multiple orders

The gap increments $g_k \equiv u_{k,i} - u_{k-1,i} \sim \mathrm{Exp}(1)$ are independent. The joint noise covariance across orders k, l at the same point is

$$\Sigma_{kl} \;=\; \mathrm{Cov}[\hat{s}_i^{(k)},\, \hat{s}_i^{(l)}] \;=\; \psi_1(\max(k,l)). \tag{66}$$

Observations at distinct points $i \neq j$ have noise-independent residuals; their covariance is pure signal: $\mathrm{Cov}[\hat{s}_i^{(k)}, \hat{s}_j^{(l)}] = K(|\mathbf{x}_i - \mathbf{x}_j|)$.

D.4 Sufficient statistic and the GLS limit

The Fisher information about λ_i from all K distances depends only on $r_{K,i}$: $L(\lambda_i) \propto \lambda_i^K \exp(-\lambda_i \frac{4\pi}{3} r_{K,i}^3)$, so GLS combination of $\hat{s}_i^{(1)}, \ldots, \hat{s}_i^{(K)}$ with the exact covariance $\mathbf{\Sigma}$ collapses to using $k = K$ alone, with noise $\psi_1(K)$. For per-point density estimation there is no gain from combining orders beyond simply using $k = K$. The self-consistency argument above provides the physical reason to prefer $k = 1$ despite this: lower noise via larger k is spurious once position uncertainty is accounted for.

D.5 Partition identity and normalisation

The raw kNN volumes do not sum to L^3. Define rescaled estimates

$$\tilde{V}_i^{(k)} \;=\; L^3 \cdot \frac{r_{k,i}^3}{\sum_{j=1}^{N} r_{k,j}^3}, \qquad \sum_{i=1}^{N} \tilde{V}_i^{(k)} = L^3. \tag{67}$$

The normalisation introduces a $\mathcal{O}(1/N)$ rank-1 correlation between observations, negligible for $N \gtrsim 10^3$.

D.6 Recovering anisotropic geometry from kNN directions

The Voronoi cell is a polyhedron. Its face normals point toward the Delaunay neighbours; its face areas weight those directions by solid-angle exposure. This directional structure is what the exact Voronoi estimator uses to reduce noise below σ_1^2, and it is what enters the Hessian estimator of Section 6. With only radial distances $r_{k,i}$ it is lost. With the full kNN *positions* $\mathbf{r}_{k,i} \equiv \mathbf{x}_{(k),i} - \mathbf{x}_i$ it is recoverable.

Shape tensor. Define the weighted displacement tensor

$$\mathbf{T}_i \;=\; \sum_{k=1}^{K} w_k \, \hat{\mathbf{n}}_{k,i} \otimes \hat{\mathbf{n}}_{k,i}, \qquad \hat{\mathbf{n}}_{k,i} = \frac{\mathbf{r}_{k,i}}{r_{k,i}}, \quad w_k \propto r_{k,i}^{-2}. \tag{68}$$

The weight $r_{k,i}^{-2}$ approximates the Voronoi face-area scaling: the face area between i and a Delaunay neighbour j scales as r_{ij}^{-2} in the isotropic limit. For a locally isotropic distribution, $\mathbf{T}_i \to \frac{1}{3}\mathbf{I}$. Anisotropy in $\mathbf{T}_i$ indicates a preferred direction in the local point distribution, corresponding to the orientation of the local density structure.

Gradient estimator. Given log-volume observations $\{\hat{s}_{(k),i}\}$ at the neighbour positions, a DTFE-analogue gradient estimator is

$$\widehat{\nabla f}(\mathbf{x}_i) \;\approx\; -\mathbf{T}_i^{-1} \sum_{k=1}^{K} w_k \, \hat{\mathbf{n}}_{k,i} \, [\hat{s}_{(k),i} - \hat{s}_i], \tag{69}$$

where $\hat{s}_{(k),i}$ is the log-volume estimate at the k-th neighbour. This is the kNN analogue of the DTFE linear interpolation over Delaunay simplices, without constructing the triangulation. For $K \geq 4$ the matrix $\mathbf{T}_i$ is generically full rank in $\mathbb{R}^3$ and the estimator is well-defined.

Approximation to the Delaunay neighbourhood. For a 3D Poisson process, the expected number of Delaunay neighbours is ≈ 15.5. With $K \sim 20$ kNN, the kNN set contains essentially all Delaunay neighbours with high probability. The difference is that equation (68) weights by $r_{k,i}^{-2}$ rather than the exact Voronoi face areas $A_{ij} \propto (r_{ij}^2 \sin^2\theta_{ij})^{-1}$, where θ_{ij} is a geometric correction from the circumcircle of the shared Delaunay facet. This is a second-order correction of order $(\nabla^2 f)\, r_{1,i}^2$, i.e. it is suppressed on scales where the smoothness approximation holds.

Decomposition of uses. The kNN data therefore supports two separable uses:

- *Density (volume) estimate*: uses only radial distances $r_{k,i}$. Self-consistency with position uncertainty selects $k = 1$; $\sigma_1^2 = \pi^2/6$ is the appropriate noise level.
- *Anisotropic / Hessian information*: uses only directions $\hat{\mathbf{n}}_{k,i}$ and inter-neighbour density contrasts. Requires $K \sim 15$–20 to span the Delaunay neighbourhood.

Neither use requires constructing the Voronoi tessellation.

D.7 Two genuine benefits of multi-order combinations

Multi-scale kernel learning. Each order k probes the density on scale $\bar{r}_k \sim (3k/4\pi\bar{n})^{1/3}$. Treating all NK observations jointly in the GP marginal likelihood with the known covariance (66) provides $N(K-1)$ additional constraints on the shape of $K(r)$ at scales between $\bar{r}_1$ and $\bar{r}_K$, beyond what a single order can provide.

Empirical Poisson diagnostic. The scatter of $\hat{s}_i^{(k)}$ across orders k at fixed i provides a per-point empirical test of the Poisson assumption. Deviations from the theoretical covariance $\mathbf{\Sigma}$ flag clustered substructure (e.g. satellite pairs, fibre-collision incompleteness, photometric-redshift catastrophes) where the model breaks down.

D.8 Information content below $\bar{r}_1$

Inter-point channel. The GP marginal likelihood constrains $K(r)$ through pair covariances at separation $|\mathbf{x}_i - \mathbf{x}_j| \approx r$. Below $r_{1,\min}$ there are no point pairs; $K(r)$ is prior-dominated regardless of k.

Intra-point channel. The difference $\hat{s}_i^{(k)} - \hat{s}_i^{(k')}$ constrains the density variance between smoothing scales $r_{k,i}$ and $r_{k',i}$. The SNR averaged over N points is

$$\mathrm{SNR} \sim \frac{\sqrt{N}\,\Delta K}{\psi_1(k) - \psi_1(k')}, \tag{70}$$

where ΔK is the kernel variance in the band. For $\Delta K \lesssim 0.2$ (weakly non-linear scales at $z = 0$) and $N = 10^5$, SNR ~ 40 at $r \sim \bar{r}_1$ but drops to zero below $\bar{r}_1$ because $\Delta K \to 0$ when both $r_{k,i}$ and $r_{k',i}$ fall below the sampling scale. The intra-point channel provides information on $K(r)$ in the interval $[\bar{r}_1, \bar{r}_K]$ only.

Hard floor. The mean nearest-neighbour distance $\bar{r}_1 \approx 0.554\,\bar{n}^{-1/3}$ is the effective resolution floor. Below $\bar{r}_1$, the GP posterior for $K(r)$ is determined by the kernel parameterisation rather than the data. A minimum scale cut at $r \approx \bar{n}^{-1/3}$ should be imposed on all inferred quantities.

Table 3: Information content on $K(r)$ vs scale ($d = \bar{n}^{-1/3}$).

Scale	Channel	Content
$r < r_{1,\min}$	None	Prior only
$r_{1,\min}$ to $0.5\,d$	Intra-point	Noise-dominated, SNR < 1
$0.5\,d$ to $2\,d$	Intra- + inter-point	Partial; few pairs
$r > 2\,d$	Inter-point	Well-constrained
$r > \bar{r}_K$	None	Prior takes over

E Hamilton-Estimator kNN Volume Estimator for Survey Data

Observational catalogs are accompanied by a random catalog that encodes the survey selection function $\phi(\mathbf{x})$, angular mask, and radial completeness. Standard two-point estimators—Hamilton (1993), Landy & Szalay (1993)—use this random catalog to cancel $\phi(\mathbf{x})$ from pair counts. Here we derive the per-point analogue: a volume estimator built from the ratio of data-to-random kNN distances that cancels $\phi(\mathbf{x}_i)$ exactly at the single-point level, requires no Voronoi construction, and is computable in $\mathcal{O}(N_D \log N_D + N_D \log N_R)$ time once the KD-trees for D and R are available.

E.1 Setup

Let $\{x_i\}_{i=1}^{N_D}$ be the data positions and $\{y_j\}_{j=1}^{N_R}$ the random positions, with $\alpha \equiv N_R/N_D$ typically in the range 10–50. The local densities are

$$n_D(\mathbf{x}) = \bar{n}_D\,(1+\delta)\,\phi(\mathbf{x}) = \bar{n}_D\,e^{f(\mathbf{x})}\,\phi(\mathbf{x}), \qquad n_R(\mathbf{x}) = \bar{n}_R\,\phi(\mathbf{x}), \tag{71}$$

where $\phi(\mathbf{x})$ is the selection function, normalised so that $\int \phi\, d^3x = V_{\text{eff}}$, and $f(\mathbf{x}) = \log \rho/\bar{\rho}$ is the log-density contrast we wish to infer.

E.2 Per-point log-density from the D/R distance ratio

The k-th nearest-data-neighbour distance $r^{(D)}_{k_D,i}$ and k-th nearest-random-neighbour distance $r^{(R)}_{k_R,i}$ satisfy, by the Gamma-distribution result of equation (62),

$$u^{(D)}_{k_D,i} \equiv n_D(\mathbf{x}_i)\,\tfrac{4\pi}{3}\big(r^{(D)}_{k_D,i}\big)^3 \;\sim\; \text{Gamma}(k_D, 1), \tag{72}$$

$$u^{(R)}_{k_R,i} \equiv n_R(\mathbf{x}_i)\,\tfrac{4\pi}{3}\big(r^{(R)}_{k_R,i}\big)^3 \;\sim\; \text{Gamma}(k_R, 1). \tag{73}$$

Taking logarithms and differencing,

$$3\log\frac{r^{(R)}_{k_R,i}}{r^{(D)}_{k_D,i}} \;=\; \underbrace{\log u^{(R)}_{k_R,i} - \log u^{(D)}_{k_D,i}}_{\epsilon_i} \;+\; \log\frac{n_D(\mathbf{x}_i)}{n_R(\mathbf{x}_i)}. \tag{74}$$

The selection function cancels in the ratio $n_D/n_R = (\bar{n}_D/\bar{n}_R)\,e^{f(\mathbf{x}_i)}$, giving the observation model. The key step is reading this equation left-to-right: the *observable* $3\log(r_{k_R}/r_{k_D}) + \log(N_R/N_D)$ on the left is a ratio of two neighbour distances, both computable from KD-tree lookups; the *signal* $f(\mathbf{x}_i)$ on the right is the log-density we want; and the *noise* ϵ_i is

the difference of two independent log-Gamma variates, whose variance $\psi_1(k_D) + \psi_1(k_R)$ is known analytically. No density estimation, no binning, no randoms subtraction in configuration space: the selection function cancels point-by-point.

$$\boxed{\hat{s}_i^{\rm H} \;\equiv\; 3\log\frac{r^{(R)}_{k_R,i}}{r^{(D)}_{k_D,i}} \;+\; \log\frac{N_R}{N_D} \;=\; f(\mathbf{x}_i) \;+\; \epsilon_i,} \tag{75}$$

where the noise is

$$\epsilon_i \;=\; \log u^{(R)}_{k_R} - \log u^{(D)}_{k_D}, \qquad \mathrm{Var}[\epsilon_i] \;=\; \psi_1(k_D) + \psi_1(k_R), \tag{76}$$

and $u^{(D)}_{k_D}$ and $u^{(R)}_{k_R}$ are independent. This is the direct per-point analogue of the Hamilton estimator: the selection function cancels at the level of individual points, not merely in ensemble averages.

E.3 Noise budget

The total noise variance $\psi_1(k_D)+\psi_1(k_R)$ has two independent contributions. The randoms term $\psi_1(k_R)$ is suppressed by increasing k_R. With $k_R = \alpha k_D$ (equal expected sphere volumes for D and R), the randoms contribution is $\psi_1(\alpha k_D)$. For $\alpha = 10$ and $k_D = 1$: $\psi_1(10) \approx 0.105$, reducing the total noise from $\psi_1(1) \approx 1.645$ to ≈ 1.75 — a minor increase. For $\alpha = 50$ and $k_D = 1$: $\psi_1(50) \approx 0.020$, so the total noise is ≈ 1.665, essentially equal to using data alone.

The randoms noise can be made negligible in practice (total noise $\lesssim 1.05\,\psi_1(k_D)$) by choosing $k_R \gtrsim 10\,\alpha\,k_D$. This choice keeps $r^{(R)}_{k_R,i} \approx r^{(D)}_{k_D,i}$, so the distance ratio directly probes the local overdensity.

E.4 Self-consistency at $k_D = 1$

The self-consistency argument of Appendix D carries over directly. The characteristic cell radius is $r^{(D)}_{1D,i}$ (nearest data neighbour), and both the density noise $\psi_1(1)$ and the position uncertainty $\sigma^2_{\rm pos} \propto |\nabla f|^2 (r^{(D)}_{1D,i})^2$ are governed by the same scale. The randoms provide $r^{(R)}_{1R,i}$ as a local estimate of the expected inter-tracer spacing under $\phi(\mathbf{x}_i)$; the ratio $r^{(R)}_{1R,i}/r^{(D)}_{1D,i}$ encodes exactly the local overdensity after selection-function correction.

E.5 Partition identity

The raw estimates $e^{\hat{s}_i^{\rm H}}$ do not sum to $V_{\rm eff}$. The effective survey volume is estimated from the randoms as $\hat{V}_{\rm eff} = N_R/\bar{n}_R$ (or equivalently, the sum of random-catalog Voronoi volumes). Normalised estimates

$$\tilde{s}_i^{\rm H} \;=\; \hat{s}_i^{\rm H} \;-\; \log\sum_{j=1}^{N_D} e^{\hat{s}_j^{\rm H}} \;+\; \log\hat{V}_{\rm eff} \tag{77}$$

satisfy $\sum_i e^{\tilde{s}_i^{\rm H}} = \hat{V}_{\rm eff}$, restoring the partition identity within the survey footprint. The subtracted term is a global constant that shifts the GP mean offset and has no effect on the inferred $K(r)$.

E.6 Recovering anisotropic geometry

Shape tensors for the data and random neighbourhoods are defined separately:

$$\mathbf{T}_i^{(D)} \;=\; \sum_{k=1}^{K_D} w_k\, \hat{\mathbf{n}}_{k,i}^{(D)} \otimes \hat{\mathbf{n}}_{k,i}^{(D)}, \qquad \mathbf{T}_i^{(R)} \;=\; \sum_{k=1}^{K_R} w_k\, \hat{\mathbf{n}}_{k,i}^{(R)} \otimes \hat{\mathbf{n}}_{k,i}^{(R)}, \tag{78}$$

with $w_k \propto r_{k,i}^{-2}$ as before. For a point far from any survey boundary, $\mathbf{T}_i^{(R)} \to \frac{1}{3}\mathbf{I}$ in the limit $K_R \to \infty$. Deviations of $\mathbf{T}_i^{(R)}$ from isotropy signal proximity to the survey edge, a hole, or a masked region.

The clustering-induced anisotropy, corrected for survey geometry, is isolated by the difference $\mathbf{T}_i^{(D)} - \mathbf{T}_i^{(R)}$. The density gradient corrected for the selection-function gradient is

$$\widehat{\nabla f}(\mathbf{x}_i) \;\approx\; -\big(\mathbf{T}_i^{(D)}\big)^{-1} \sum_{k=1}^{K_D} w_k\, \hat{\mathbf{n}}_{k,i}^{(D)} \big[\hat{s}_{(k),i}^{(D)} - \hat{s}_i^{(D)}\big] \;+\; \big(\mathbf{T}_i^{(R)}\big)^{-1} \sum_{k=1}^{K_R} w_k\, \hat{\mathbf{n}}_{k,i}^{(R)} \big[\hat{s}_{(k),i}^{(R)} - \hat{s}_i^{(R)}\big], \tag{79}$$

where the second term subtracts the selection-function gradient estimated from the randoms. This is the per-point, DTFE-analogue gradient estimator for a masked, inhomogeneous survey.

E.7 KD-tree implementation

The Hamilton kNN estimator requires only two per-point neighbour lookups after the trees are built, which the two-point estimator already requires. Algorithm E.7 gives the pseudocode.

Algorithm 1: Hamilton kNN volume estimate

Input: data positions $D = \{x_i\}$, random positions $R = \{y_j\}$, orders k_D, k_R
Output: per-point log-density estimates $\{\hat{s}_i^{\mathrm{H}}\}$

1. Build KD-tree $\mathcal{T}_D$ from D and $\mathcal{T}_R$ from R.
2. For each $x_i \in D$:
   (a) Query $\mathcal{T}_D \setminus \{x_i\}$ for k_D-th nearest data neighbour $\to r_{k_D,i}^{(D)}$ and directions $\{\hat{\mathbf{n}}_{k,i}^{(D)}\}_{k=1}^{K_D}$
   (b) Query $\mathcal{T}_R$ for k_R-th nearest random neighbour $\to r_{k_R,i}^{(R)}$ and directions $\{\hat{\mathbf{n}}_{k,i}^{(R)}\}_{k=1}^{K_R}$
   (c) Set $\hat{s}_i^{\mathrm{H}} = 3\log(r_{k_R,i}^{(R)}/r_{k_D,i}^{(D)}) + \log(N_R/N_D)$
3. Apply normalisation (77) using $\hat{V}_{\mathrm{eff}} = N_R/\bar{n}_R$.
4. *(Optional)* Compute shape tensors $\mathbf{T}_i^{(D)}, \mathbf{T}_i^{(R)}$ and gradient estimate (79) from the stored directions.

Cost. Building each KD-tree is $\mathcal{O}(N \log N)$. Each query for k neighbours costs $\mathcal{O}(k \log N)$; the full pass over N_D points is $\mathcal{O}(N_D k_D \log N_D + N_D k_R \log N_R)$. For $k_D = k_R = 20$, $N_D = 10^6$, $\alpha = 20$: this is $\sim 20 \times 10^6 \times 20 \approx 4 \times 10^8$ operations — comfortably sub-minute on a single core, and trivially parallelisable. The KD-tree construction dominates in practice and is shared with any subsequent two-point estimator, so the incremental cost of the per-point density estimate is essentially free.

E.8 Relation to the standard Hamilton estimator

The standard Hamilton (1993) estimator accumulates pair counts in separation bins:

$$\hat{\xi}(r) \;=\; \frac{DD(r) \cdot RR(r)}{[DR(r)]^2} - 1. \tag{80}$$

Our per-point estimator is the *field-level* generalisation. To see the connection explicitly, note that $DD(r)\,dr$ counts pairs at separation r in the data catalog; at the per-point level this is the number of data neighbours with $r_{k,i}^{(D)} \approx r$, which encodes the local density. The ratio $r_{k_R,i}^{(R)}/r_{k_D,i}^{(D)}$ is the per-point, per-scale analogue of $\sqrt{RR/DD}$. Taking the log and squaring,

$$e^{2\hat{s}_i^{\mathrm{H}}} \;\approx\; \frac{(r_{k_R,i}^{(R)})^6}{(r_{k_D,i}^{(D)})^6} \cdot \frac{N_D^2}{N_R^2} \;\sim\; \frac{n_D(\mathbf{x}_i)^2}{n_R(\mathbf{x}_i)^2} \cdot (\text{noise}), \tag{81}$$

which is the local version of the DD/RR ratio inside equation (80). The Hamilton estimator then arises from averaging these local ratios over all pairs in a bin, while our GP marginal likelihood uses all the per-point observations jointly to infer the smooth field $f(\mathbf{x})$ and its correlation function simultaneously.

The advantages over the standard Hamilton estimator are the same as those discussed in Section 3: full posterior covariance on $\xi_\delta(r)$, joint field reconstruction, and anisotropic (Hessian/web) information, all at the cost of the log-normal model assumption. The survey-geometry qualification of Section 3 is automatically handled here: the randoms encode $\phi(\mathbf{x})$ and the cancellation is exact per point, so no additional mask treatment is required beyond having a good random catalog.

E.9 Angular confidence from the random shape tensor

The gradient and Hessian estimates of equations (79) and Section 6 require inverting $\mathbf{T}_i^{(R)}$. Near a survey edge, hole, or masked region, directions pointing into the missing solid angle have no random neighbours; the corresponding eigenvalue of $\mathbf{T}_i^{(R)}$ shrinks toward zero and the inverse amplifies noise in that direction. Since $\mathbf{T}_i^{(R)}$ is built from random neighbour *directions alone* — density values never enter it — this degradation is a purely geometric quantity, computable as a free by-product of the KD-tree query.

Directional confidence. For a complete, isotropic random distribution, $\mathbf{T}_i^{(R)} \to \frac{1}{3}\mathbf{I}$. The gradient estimator amplifies noise in direction $\hat{\mathbf{e}}$ by the factor $\hat{\mathbf{e}}^T\,(\mathbf{T}_i^{(R)})^{-1}\,\hat{\mathbf{e}}$ relative to the isotropic baseline 3. The directional confidence is defined as

$$\mathrm{Conf}_i(\hat{\mathbf{e}}) \;\equiv\; \frac{1/3}{\hat{\mathbf{e}}^T\,(\mathbf{T}_i^{(R)})^{-1}\,\hat{\mathbf{e}}} \;\in\; [0,1], \tag{82}$$

which equals unity for isotropic coverage and degrades toward zero in directions absent from the survey. The eigenvector of the minimum eigenvalue $\lambda_{\min}(\mathbf{T}_i^{(R)})$ identifies the direction of greatest uncertainty; for a point near a planar survey boundary this is the outward normal to the boundary.

Scalar summary per point. Two scalar summaries are useful:

$$Q_i \;\equiv\; 3\,\lambda_{\min}(\mathbf{T}_i^{(R)}) \;\in\; [0,1], \tag{83}$$

$$\mathcal{Q}_i \;\equiv\; 27\,\det(\mathbf{T}_i^{(R)}) \;=\; \prod_{a=1}^{3} 3\lambda_a^{(R)} \;\in\; [0,1]. \tag{84}$$

Q_i measures worst-case degradation: it is the ratio of the minimum eigenvalue to the isotropic value 1/3, and the gradient estimator is reliable in all directions only when $Q_i \approx 1$. $\mathcal{Q}_i$ is the geometric mean of the three directional confidences, equal to the volume ratio of the coverage ellipsoid to the isotropic sphere. Near a planar edge, one eigenvalue vanishes: $Q_i \to 0$ and $\mathcal{Q}_i \to 0$. Near a corner, two eigenvalues vanish and both measures collapse.

Condition number. The noise amplification in the worst direction is the condition number

$$\kappa_i \;=\; \frac{\lambda_{\max}(\mathbf{T}_i^{(R)})}{\lambda_{\min}(\mathbf{T}_i^{(R)})}, \tag{85}$$

which gives the ratio of the uncertainty on the gradient component in the least-covered direction to that in the best-covered direction. A practical threshold of $\kappa_i < 10$ (corresponding to $Q_i > 0.05$) is sufficient to flag unreliable gradient estimates near boundaries.

Interpretation and use. All three quantities—$\mathrm{Conf}_i(\hat{\mathbf{e}})$, Q_i, $\mathcal{Q}_i$, and κ_i—are computed from the random neighbour directions $\{\hat{\mathbf{n}}_{k,i}^{(R)}\}$ alone. No density values, no model, no separate boundary-detection step is required. The confidence degrades continuously from unity in the survey interior to zero at the boundary, tracing the angular completeness of the random catalog at each point. This provides:

1. A per-point weight for the GP observation: points with low Q_i contribute less to the inference of the density gradient and Hessian eigenvalues, without requiring explicit boundary masking.

2. A diagnostic for the web-type posterior (Section 6): $\kappa_i \gg 1$ indicates that the T-web classification at point i is boundary-contaminated and should be downweighted or excluded.

3. A quality flag for the survey analysis: the map of $\mathcal{Q}_i$ over the footprint directly visualises the effective depth of the random catalog and identifies regions requiring additional randoms or mask treatment.

Note that Q_i and $\mathcal{Q}_i$ depend on K_R through the finite sampling of the sphere by random neighbours. In the interior, $\mathbf{T}_i^{(R)}$ converges to $\frac{1}{3}\mathbf{I}$ as $K_R \to \infty$, so $Q_i \to 1$ and $\kappa_i \to 1$ with corrections of order $K_R^{-1/2}$. Near a boundary, convergence is slower because the missing solid angle is a fixed geometric deficit independent of K_R. A practical choice is $K_R \sim 20\alpha$ (twenty times the random oversampling ratio) to make sampling noise in $\mathbf{T}_i^{(R)}$ negligible relative to genuine geometric incompleteness.

F Anisotropic Volume Correction from the LPT Jacobian Prior

The Voronoi volume estimator assigns volume isotropically: each cell is the Dirichlet domain of the Euclidean metric. In an isotropic Poisson point process this is unbiased, giving $\sigma_V^2 \approx 0.289$. However, cosmological density fields are not isotropically distributed: the Lagrangian volume element at a point in a filament or sheet is a strongly anisotropic deformed tetrahedron, collapsed along the caustic directions and extended along the void-facing direction. The Voronoi cell, finding few neighbours in the uncollapsed direction, bleeds volume there. This systematic is visible in Abel et al. [2012] (their Fig. 13) as a density-dependent bias of the zeroth-order Voronoi estimate relative to the sheet density in the range $\rho/\bar{\rho} \sim 3$–30, precisely the single-stream sheet and filament regime where one or two Lagrangian eigenvalues approach unity while the third has not. The bias vanishes in deep voids (isotropic expansion) and is superseded by multi-stream complexity in collapsed halos.

F.1 LPT eigenvalue distribution

In Lagrangian perturbation theory, the mapping from initial position $\mathbf{q}$ to Eulerian position $\mathbf{x}$ has Jacobian

$$J(\mathbf{q}) \;=\; (1-\lambda_1 D)(1-\lambda_2 D)(1-\lambda_3 D), \tag{86}$$

where $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are eigenvalues of the deformation tensor $\psi_{ab} = \partial^2\Phi_0/\partial q_a \partial q_b$ (proportional to the initial tidal tensor) and D is the linear growth factor. For a Gaussian random field with linear variance σ^2, the joint distribution of eigenvalues is the Doroshkevich [1970] distribution:

$$p(\lambda_1, \lambda_2, \lambda_3) \;\propto\; \prod_{a<b}(\lambda_a - \lambda_b)\;\exp\!\left(-\frac{3I_1^2 - 15I_2}{2\sigma^2}\right), \tag{87}$$

where $I_1 = \sum_a \lambda_a$ and $I_2 = \sum_{a<b} \lambda_a \lambda_b$.

Conditioned on the Eulerian log-density $f_i = -\log J(\mathbf{q}_i)$, equation (87) gives a non-trivial posterior on the eigenvalue ratios λ_a/I_1. The *shape* of the Lagrangian volume element — which determines the Voronoi volume bias — is set by precisely these ratios. In a sheet ($\lambda_1 D \approx \lambda_2 D \approx 1$, $\lambda_3 D \ll 1$), the volume element is a thin slab; in a filament ($\lambda_1 D \approx 1$, $\lambda_2 D \approx \lambda_3 D \ll 1$), it is a needle. The Voronoi cell cannot represent either shape isotropically, and the bias of $\log V_i$ as an estimator of $-f_i$ acquires a systematic component.

F.2 Anisotropy-dependent bias and noise

Let $\boldsymbol{\lambda}_i \equiv (\lambda_1, \lambda_2, \lambda_3)$ be the deformation eigenvalues at point i. The observation model of equation (5) generalises to

$$s_i \;=\; -f_i \;-\; c_V(\boldsymbol{\lambda}_i) \;+\; \epsilon_i, \qquad \mathrm{Var}[\epsilon_i] \;=\; \sigma_V^2(\boldsymbol{\lambda}_i), \tag{88}$$

where the mean offset $c_V(\boldsymbol{\lambda})$ and noise variance $\sigma_V^2(\boldsymbol{\lambda})$ are functions of the local eigenvalue configuration. In the isotropic limit ($\lambda_1 = \lambda_2 = \lambda_3$), these reduce to the Poisson–Voronoi values of Appendix A. In highly anisotropic configurations, the mean offset is

larger (Voronoi overestimates V_i, underestimates ρ_i) and the variance increases as the uncollapsed direction is poorly sampled by neighbours.

The functions $c_V(\boldsymbol{\lambda})$ and $\sigma_V^2(\boldsymbol{\lambda})$ can be tabulated from log-normal mocks with prescribed deformation tensor statistics (Section 7) or approximated analytically. To leading order in the anisotropy parameters $\eta_a \equiv \lambda_a/I_1 - 1/3$,

$$c_V(\boldsymbol{\lambda}) \approx c_V^{(0)} + c_V^{(2)} \sum_a \eta_a^2, \tag{89}$$

$$\sigma_V^2(\boldsymbol{\lambda}) \approx \sigma_V^{2,(0)} + \sigma_V^{2,(2)} \sum_a \eta_a^2, \tag{90}$$

where $c_V^{(0)}$ and $\sigma_V^{2,(0)}$ are the isotropic values, and the quadratic coefficients $c_V^{(2)}$, $\sigma_V^{2,(2)}$ are calibrated from simulations. The leading correction vanishes by symmetry for $\eta_a \to 0$ and grows quadratically in the anisotropy. In the filament regime $\sum_a \eta_a^2 \sim 2/3$, which gives a correction of order 20–30% to c_V, consistent with the bias seen in Abel et al. [2012].

F.3 Estimating the deformation eigenvalues

The eigenvalues $\boldsymbol{\lambda}_i$ are not directly observable but can be estimated from two sources within the GP-$\log\rho$ framework:

GP Hessian of f. The posterior Hessian $\mathbf{H}_i^f = \partial_a\partial_b f(\mathbf{x}_i)$ from Section 6 is related to the deformation tensor eigenvalues in the single-stream regime. In LPT, $\partial_a\partial_b f \approx -\partial_a\partial_b \log J$, so the eigenvalues of $\mathbf{H}_i^f$ are monotonically related to $\boldsymbol{\lambda}_i D$ through the chain rule.

kNN shape tensor. The shape tensor $\mathbf{T}_i^{(D)}$ from equation (68) gives an empirical, model-free estimate of the local deformation axis ratios. Its eigenvalues $\mu_1 \geq \mu_2 \geq \mu_3$ measure the solid-angle-weighted distribution of neighbour directions; their deviation from isotropy (1/3) traces the collapse anisotropy. The anisotropy parameter $\sum_a \eta_a^2$ entering equations (89)–(90) can be estimated directly from $\mathbf{T}_i^{(D)}$ without any LPT model assumption.

Doroshkevich prior. The GP posterior on f_i and the local Hessian constrain $I_1 = \sum \lambda_a$ and $J = (1 - \lambda_1 D)\cdots$. Combined with the Doroshkevich distribution (87) as a prior, this gives a posterior on the full eigenvalue triplet $(\lambda_1, \lambda_2, \lambda_3)$ at each point. The expectation of $\sum_a \eta_a^2$ under this posterior is the correction to the noise model. At 2LPT order, the Jacobian acquires the correction $J^{(2)} = J^{(1)} + D_2 \sum_{a<b} \lambda_a\lambda_b$, extending the validity of the prior from the linear to the mildly non-linear regime ($\delta \lesssim$ few) that spans precisely the biased range $\rho/\bar{\rho} \sim$ 3–30.

Degeneracy with tidal bias and its resolution. The leading anisotropy correction $\sum_a \eta_a^2 \propto s^2/\sigma^2$ is proportional to the same tidal operator that enters the Lagrangian bias expansion at second order (Section 5.6). Both shift s_i and $\tilde{T}_{i,ab}$ along the same local tidal axis, and are degenerate in both channels. The separation rests on the fact that $\gamma_V(\boldsymbol{\lambda}_i)$ is a *deterministic* output of the GP posterior Hessian with no free parameters, while $b_{s^2}^L$ is a free bias coefficient: the fixed-point iteration of Section 5.6 evaluates γ_V from the

GP Hessian at each step, subtracts it from the shape-tensor residual, and attributes the remainder to $b^L_{s^2}$. Note also that neither effect contributes to the quadrupole $K_2^{(s)}(r)$ of the redshift-space kernel (their axes are random), so $f_{\rm growth}$ inferred from the quadrupole is free of this degeneracy.

F.4 Fixed-point iteration

The anisotropy correction is self-consistent: the corrected noise model depends on the GP posterior Hessian, which depends on the noise model. The same fixed-point structure as Appendix C applies:

1. Initialise with isotropic noise $c_{V,i} = c_V^{(0)}$, $\sigma^2_{V,i} = \sigma_V^{2,(0)}$.
2. Compute GP posterior $\to$ Hessian of f $\to$ deformation axis ratios $\hat{\boldsymbol{\eta}}_i$.
3. Update $c_{V,i}$ and $\sigma^2_{V,i}$ via equations (89)–(90).
4. Update observations: $s_i \to s_i - (c_{V,i} - c_V^{(0)})$.
5. Recompute GP posterior with heteroscedastic noise $\{\sigma^2_{V,i}\}$.
6. Repeat until convergence (typically 3–5 iterations in the linear regime; more in the mildly non-linear filament regime).

In deep voids and the linear regime, $\hat{\boldsymbol{\eta}}_i \approx 0$ and the correction is negligible. In sheet and filament environments, the correction shifts the mean offset by ~ 0.1–0.3 and increases the per-point noise, correctly reducing the weight given to Voronoi volumes in anisotropically deformed regions.

F.5 Relation to the kNN estimator

The kNN estimator (Appendix D) with $k = 1$ avoids the Voronoi anisotropy bias by construction: the sphere of radius $r_{1,i}$ is isotropic, and $\sigma_1^2 = \psi_1(1)$ is independent of the local deformation. The bias shown in Fig. 13 of Abel et al. [2012] is precisely a Voronoi artifact that the kNN estimator does not share. The trade-off is the larger noise $\sigma_1^2 \approx 1.645$ versus $\sigma_V^2 \approx 0.289$ in the isotropic limit; in the anisotropic filament regime, the corrected Voronoi noise $\sigma_V^2(\boldsymbol{\lambda})$ grows toward σ_1^2, reducing the advantage of the Voronoi estimator over kNN.

The anisotropic correction and the kNN estimator therefore bracket the same physics from opposite directions: the Voronoi estimator is more precise in isotropic environments and requires a model-dependent correction in anisotropic ones; the kNN estimator is uniformly noisier but uniformly unbiased. A natural hybrid uses the kNN estimator as a prior on f_i in the first GP iteration, then switches to the Voronoi estimator (with anisotropy correction) in subsequent iterations once the local deformation is estimated.

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